JANSETU / EWA HEALTHCARE

Business operations and marketplace roadmap

Daily in-house sales, services, billing and inventory.
Optional patient marketplace.

Prepared 9 November 2026 | Branch: pharmyx-fixes

Source commit: `9ca8b54606581f2158e760cad79aa7b845529184`

SOURCE CROSS-CHECK	FINDING
70 bounded capabilities	19 existing / 18 partial / 30 missing / 3 unverified
Scope	Six requested service areas; all 53 modules inventoried
Business model	Private pharmacy, lab, clinic and hospital operations; public listing is optional
Confirmed first release	Connected counter sales; hospital OPD and billing first
Lab partner model	Provider APIs where available; Jansetu panel for labs without APIs
Phone access	Basic-phone doctor helpline and partner-assisted top-ups; F14 separately estimated after M1/M2
Indicative rollout	M0/M1 to re-estimate; provisional pilot post-M0 weeks 11-12; rollout weeks 19-24. Rebase dates when gates or added scope extend.
Throughput target	100 aggregate incoming HTTP requests/s; not yet benchmarked

Existing means source implementation was identified for the named backend capability. It does not establish a finished client, secure deployment or passing runtime test. Counts are not percentage-complete scores.

Application code was not changed. The current lockfile mismatch blocks clean installation and runtime verification. Security blockers must be resolved before launch.

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1. Existing, partial and missing capabilities

Source: documentation/planning/CAPABILITY_MATRIX.md

Second source review: 8 November 2026. **Baseline:** `pharmyx-fixes`, commit `9ca8b54606581f2158e760cad79aa7b845529184`. No application changes occurred between the first review and this cross-check.

This document answers whether the requested capabilities are present in this backend. It does not equate a database field, helper, API or status label with a complete working product. Existing source has not passed a runtime verification in this environment: clean install remains blocked by the committed lockfile mismatch.

The expanded requirement adds daily in-house business operations and inventory for labs, clinics/doctors and hospitals as well as pharmacies. Nine additional bounded checks are recorded in section H; the original 60 remain as the earlier baseline. The earlier expanded baseline had 69 checks. The new basic-phone requirement adds C11, making 70 checks: 19 Existing, 18 Partial, 30 Missing and 3 Unverified. Counts describe the chosen requirements, not a completion percentage. See BUSINESS_OPERATIONS.md.

Status definitions

* **Existing:** a backend route/use case and supporting source were identified for the specifically named capability. Reuse and test it; this does not certify security or production correctness.
* **Partial:** useful code or data exists, but the named workflow has material gaps.
* **Missing:** no dedicated implementation of the named workflow was identified after graph discovery, source/route review and targeted fallback searches. This is a finding about this checkout, not every EWA/client repository.
* **Unverified:** cannot be established from this checkout without client code, deployment evidence or execution.

Statuses apply to the bounded capability in each row. They are not completion percentages. Delivery references M0-M8 point to the roadmap.

Six-area result

Requested area	Result	What can be reused / what is missing
App pharmacy and in-house sales	**Partial**	Catalogue, stock, cart, orders, counter-sale data and manual collection exist. Pharmacist review, complete POS interface, sales invoices, shifts and returns need work.
Retail and wholesale through app/counter	**Partial**	Both price fields and buyer/channel routing exist. Full price-list policy, purchasing, UOM conversion, trade credit and accounting documents are missing.
Hospital HMS	Missing dedicated HMS	In-person visits/documents are reusable. Facility OPD, tariffs/billing and central/department supply stores/consumption are missing; basic stores are initial scope.
Lab management	**Partial**	API booking/report/callback services exist. The missing panel now includes direct lab customers, invoices/receipts, consumable stock and signatory/review, as well as app assignments.
Doctor dashboard	Partial product; existing portal APIs	Profile/slot/appointment/status/prescription/history/stats APIs exist. Shared EHR, queue, business billing, clinic supplies and complete UI need work.
Third-party vendor dashboard	Missing general workspace	Partner CRUD/adapters/scoped roles are foundations. Standalone business/module/publication controls, staff, private catalogue, stock and commercial self-service remain incomplete.

Confirmed lab delivery model

Labs are third-party vendors with two delivery routes: **API-connected third-party labs** use provider adapters; **panel-managed third-party labs** without APIs use the Jansetu lab partner panel. The existing API adapters remain reuse scope. The missing deliverable is the lab-specific staff/order/report panel and its backend permissions, not a requirement that every partner install a full laboratory information system.

Both routes share patient/order identifiers, assigned lab/branch scope, sample and report status, original/corrected report versions, clinician-review tasks, payments and audit. Panel report release triggers the same notifications/events and optional outgoing webhooks as an API-supplied report. A lab-owned API or webhook endpoint is not needed for the panel route. API-to-panel handover must be explicit and auditable to avoid duplicate orders or competing status updates. The minimum panel ships in M4; broader vendor self-service expands in M5.

The panel must also support a lab's own unlisted walk-in/referral business, private billing and basic reagent/consumable purchasing/usage. API availability does not determine marketplace listing or stock ownership; choose one authoritative writer for each resource.

A. Pharmacy ordering, inventory and counter sales

ID	Capability	Status	Verified scope and remaining work
A01	Pharmacy onboarding and verification data	**Existing**	Pharmacy profile, document/licence fields and active/verified operational guards exist. Add individual staff, branch/licence expiry and verification history. Evidence SRC01.
A02	Medicine catalogue and discovery	**Existing**	Medicine models, search/nearby APIs and prescription/tax flags exist. Master-data governance and responsive client experience still need validation. SRC02.
A03	Patient/pharmacy app cart and checkout	**Existing**	Actor-scoped cart, current-price refresh/conflict and order creation exist; pharmacy buyers cannot buy from themselves. SRC03.
A04	Batch FEFO stock and order reservation	**Existing**	Batch expiry/status filter, ordered allocation, reservation/consumption/release and transactional order creation exist. Concurrency tests must run on a replica set. SRC04.
A05	Low-stock query and adjustment history	**Existing**	`/pharmacy-inventory/low-stock` and `/adjustments`, reorder threshold and adjustment records exist. This proves a worklist/query, not automated replenishment messages or purchasing. SRC04.
A06	Inventory spreadsheet import/retry	**Existing**	Preview, commit, failed-row download, retry and cleanup exist. Large imports need bounded worker execution and real recovery tests. SRC05.
A07	Counter sale backend and cash/COD marking	**Existing**	`in_store`, `isWalkIn`, customer/buyer references, `markPaid`, actor checks and conditional paid-state update exist. Complete cashier shifts and financial receipt workflow do not. SRC06.
A08	Prescription review before dispensing	**Partial**	Prescription URLs, required-medicine check and pending/approved/rejected enum exist. No exposed pharmacist review/release workflow was identified. SRC06.
A09	Pickup/delivery operations	**Partial**	Pickup/delivery modes and packed/ready/out-for-delivery/completed states exist. Dispatch assignment, proof of delivery and service-level exceptions are not established. SRC06.
A10	Recurring medicine and refill reminders	**Partial**	Subscriptions, next-refill dates, reminder channel, pause/update and order reconciliation exist. No due-date refill dispatch scheduler/consumer was identified; booking reminders are different. SRC07.
A11	Completed-sale return, batch restock and credit	**Partial**	Generic `returnStock`/`atomicReturn` helpers and adjustment reason `return` exist. `atomicReturn` increments aggregate available quantity; a customer-return workflow that restores the correct batch, issues credit and reconciles refund was not identified. SRC04/SRC06.
A12	Cashier register/shift close and sales invoice	**Missing**	No dedicated register/shift or pharmacy sales invoice/credit-note workflow found. Order numbers and specialist payout invoices do not supply these functions. SRC06/SRC12.

B. Retail, wholesale and commercial backoffice

ID	Capability	Status	Verified scope and remaining work
B01	Retail and wholesale prices	**Existing**	Inventory stores both; `resolveOrderItemsPricing` selects using the buyer pharmacy reference and has fallback behavior. Batch prices are stored, but checkout selects top-level inventory prices. SRC04/SRC06.
B02	App and counter sales channels	**Existing**	`orderType` supports online/in-store separately from customer/pharmacy buyer. Do not rebuild the basic channel split. SRC06.
B03	Server pricing and override governance	**Partial**	App checkout passes no client unit price and disables overrides. Counter create can allow unit-price overrides; approval/audit policy and versioned price-list entitlements need work. SRC03/SRC06.
B04	Discounts, tax snapshots and paise totals	**Existing**	Order item discount, HSN/GST and minor-unit amount fields/calculations exist. Finance must validate rounding, tax interpretation and invoice treatment. SRC06.
B05	Pack/strip/unit conversion	**Missing**	Current stock uses numeric quantities; no explicit stock/sale UOM conversion model or conversion workflow identified. SRC04/SRC06.
B06	Purchase order, GRN and supplier payable	**Missing**	Purchase price and `purchase` adjustment reason exist, but no supplier/PO/GRN/vendor-invoice chain was found. SRC04.
B07	Wholesale credit limit and receivable aging	**Missing**	Pending/partially-paid labels and wallets are not a trade-credit facility. No credit approval/limit/aging workflow found. SRC06/SRC11.
B08	Customer price lists, slabs and sales credit notes	**Missing**	No dedicated effective-dated buyer price-list/slab or linked sales credit-note implementation identified. Extend existing price routing. SRC06.

C. Doctor and consultation management

ID	Capability	Status	Verified scope and remaining work
C01	Doctor self profile, slots, appointments and stats	**Existing**	Specialist `me` routes cover profile, slots, day/month/all appointments, status, history, prescription and stats. SRC08.
C02	In-person and assisted booking foundation	**Partial**	Consultation mode includes in-person; pharmacy/admin actors can create bookings for a user. Dedicated reception/walk-in queue, triage and clinician intake workflow are not implemented. SRC09.
C03	Prescription medicine fields and PDF	**Existing**	Name, dosage, frequency, duration, instructions, diagnosis, advice, tests advised and generated PDF already exist. Separate strength/route fields, finalization and immutable amendments remain to add. SRC08/SRC09.
C04	Patient consultation history	**Partial**	Specialist-local consultation history and pharmacy-linked summaries exist. They are not a consent-controlled longitudinal EHR across clinicians, labs and hospitals. SRC08/SRC15.
C05	Free-follow-up rule and appointment reminders	**Existing**	One-level benefit rules and scheduled booking reminders exist. Preserve the benefit rule while testing failure recovery; general care task follow-up is separate. SRC09/SRC14.
C06	Consultation collection/payment integration	**Existing**	Fee, coupon/wallet and gateway-payment flows are implemented. M0 now gates atomic customer-wallet effects (R17), resume-before-retry, captured-only webhook/reconciliation and safe late/cancelled-payment recovery; receipts/cash and approved liability mappings expand later. Integration existence is not collection correctness. SRC09/SRC11.
C07	Specialist payout accounting	**Existing**	Admin APIs calculate billings, record partial/full disbursals and generate invoice/receipt assets. They do not execute or confirm a bank transfer. Doctor-accessible settlement self-service is not supplied by admin-only APIs. SRC12.
C08	Atomic shared app/walk-in capacity	**Partial**	Capacity fields/availability checks exist, but booking save and slot increment are separate; increment lacks a capacity condition. Must fix before concurrent expansion. SRC09.
C09	Clinical sign-off, amendment and report-review inbox	**Missing**	Prescriptions update the current booking object; no dedicated immutable clinical-version and clinician report-review task workflow found. SRC08/SRC09.
C10	Finished doctor application UI	**Unverified**	`doctor_app` contains API documentation, not dashboard source. Other client repositories were not reviewed in this checkout.
C11	Basic-phone doctor helpline and partner-assisted patient top-ups	**Missing**	Existing recharge and unrouted PharmacyWalletService.payConsultation are reuse primitives: pharmacy linkage, transactional debit and conditional booking-paid guard exist. The latter trusts req.amount and lacks a dedicated validated/idempotent route contract. No complete partner-to-patient top-up, lab/hospital wallet or IVR bridge/clinical-charge workflow is established. F14/P1 select Partner-paid consultation, Consultation-only balance or General patient wallet; M1 identity remains required.

Review update: AR-01-AR-36 (`documentation/planning/ARCHITECTURE_REVIEW.md`) adds payment/report/refund-budget and enforcement precision. C11 remains a Missing end-to-end requirement despite the newly acknowledged sponsorship primitive; the 70-row counts do not change. AR-21 corrects the earlier incomplete reuse description, not the live capability status. Phone specification.

D. Laboratory orders, reports and webhooks

ID	Capability	Status	Verified scope and remaining work
D01	API-connected third-party lab booking	Existing	LabStack/Healthians integration, catalogue/pricing and booking/payment/cancel/retry/report paths exist. These cover the API-capable route; real provider behavior remains untested here. SRC10.
D02	Report retrieval/storage and generic file upload	Existing	External report retrieval/persistence and booking report/file structures exist. Preserve retrieval but fix stable-name overwrite/public cache in M0. Partial/final and corrected bytes need immutable versions; legacy overwritten history is unknown. SRC09/SRC10/SRC13.
D03	Incoming booking-status callback processing	Partial	Callback routes, payload validation, raw storage and booking/report sync exist. Provider authentication, durable dedupe/order checks and processing isolation are incomplete. SRC10.
D04	Outgoing webhook subscriptions and delivery	Missing	No general subscriber/secret/event delivery-attempt/replay workspace identified. Existing incoming callbacks are not an outbound event platform. SRC10/SRC14.
D05	Partner panel and direct lab operations	Missing	No dedicated lab staff/order/sample/report workbench found. Expand the panel to direct/private test billing and consumable stock using shared business scope; existing file/provider services are foundations. Full analyzer/LIS is not a prerequisite. SRC10/SRC16/SRC19.
D06	Signatory verification and report amendments	Missing	Upload/storage does not supply signatory release, original/version linkage and reasoned corrected report workflow. SRC09/SRC10/SRC13.
D07	Critical result escalation and acknowledgment	Missing	Generic notification delivery does not establish named clinical escalation, direct communication and acknowledgment. SRC14/SRC16.
D08	Optional analyzer/QC/instrument workflows	Missing	Full LIS/instrument automation is absent and optional; basic reagent/consumable stock belongs to initial daily operations scope, covered in section H. SRC16/SRC19.

E. Hospital management

ID	Capability	Status	Verified scope and remaining work
H01	In-person visit and external documents	Partial	Reuse consultation mode, clinicians, booking files/reports and payment foundations. A specialist's hospital/clinic text field is not a hospital organization. SRC09/SRC13.
H02	Hospital branches, departments and staff	Missing	No dedicated facility/department/staff-membership model/workflow identified. SRC16.
H03	OPD registration/queue, tariffs and billing	Missing	Appointment creation is not a facility OPD workflow. No dedicated queue, tariff, hospital invoice/cash register implementation found. SRC09/SRC16.
H04	Clinical referral, acceptance and discharge	Missing	Generic documents can store a discharge file. No dedicated clinical referral/acceptance/outcome workflow found; the existing referral module awards acquisition rewards. SRC15.
H05	IPD, beds, nursing and OT	Missing	No dedicated admission/transfer/bed/ward/nursing/OT modules found. User approved these as later scope. SRC16.

F. General vendor self-service

ID	Capability	Status	Verified scope and remaining work
F01	Partner directory and provider adapters	**Existing**	Partner type/profile/verification fields and external provider clients exist. SRC10/SRC15.
F02	Scoped specialist/pharmacy actor patterns	**Partial**	Specific roles and own pharmacy/specialist guards exist. General vendor organization, branch and individual staff memberships are missing. SRC01/SRC08/SRC16.
F03	Vendor staff/catalogue/capacity/ assigned-service workspace	**Missing**	No general vendor login/staff/assignment/self-service domain found. The lab-specific panel is an M4 slice of this shared partner model; broader self-service follows in M5. SRC15/SRC16.
F04	Vendor statements, disputes and settlements	**Missing**	Specialist admin billing and pharmacy wallets are reusable but do not supply a general vendor settlement/dispute portal. SRC11/SRC12/SRC16.
F05	Scoped third-party API clients and webhook settings	**Missing**	Provider integration credentials do not establish a customer-facing API-client scope/revocation/quota/subscription model. SRC10/SRC16.

G. Shared patient, security and platform capabilities

ID	Capability	Status	Verified scope and remaining work
G01	User, dependent and pharmacy-customer records	Existing	Account/dependent profiles and pharmacy relationships exist. They are the starting point for identity migration, not a complete master-patient implementation. SRC15.
G02	Permanent patient ID, no-phone assisted identity and merge	Missing	No dedicated permanent Jansetu patient key, authorized duplicate-merge history or no-phone assisted identification workflow found. Pharmacy customer creation currently matches by phone. SRC15.
G03	Versioned consent, withdrawal and caregiver authority	Missing	No dedicated clinical/privacy consent-event and representative-authorization implementation found. OAuth consent is unrelated. SRC16.
G04	Complete clinical EHR/episodes and care timeline	Partial	Booking clinical fields, reports, health checks and scoped summaries exist. Cross-service encounter/provenance/version/consent and outstanding-task linkage is missing. SRC08/SRC09/SRC15.
G05	Follow-up, clinical emergency and grievance cases	Missing	Appointment reminders/refill preferences/reward referrals exist, but no shared care-task/clinical emergency/grievance case domain identified. SRC14/SRC15/SRC16.
G06	Authentication and role/resource enforcement	Partial	JWT and selected role/HR/pharmacy guards exist. OTP bypass, fallback secret and user object-access gaps remain release blockers. SRC16/SRC17.
G07	Clinical file authorization	Partial	Upload/download APIs exist; public static clinical-file access and caching do not meet the proposed private-record boundary. SRC13/SRC17.
G08	Persisted jobs and notifications	Partial	Mongo jobs, booking reminders and order side-effect consumers exist. Leases/retry/dead letters/outbox/worker draining need hardening; refill dispatch is not wired. SRC07/SRC14.
G09	Record-access audit, retention and privileged lifecycle	Missing	Transaction/adjustment ledgers and ordinary logger exist. No comprehensive purpose-based read/export/clinical-amendment audit and retention/hold workflow found. SRC16/SRC17.
G10	ABDM/ABHA health information exchange	Missing	No dedicated consent-driven ABDM exchange implementation found. Optional future integration must not be inferred from ordinary patient/report fields. SRC16.
G11	100 RPS, production encryption, backup/restore and UI completion	Unverified	No benchmark/restore/deployment evidence or full clients were exercised. Their absence from this checkout is not evidence that no external infrastructure exists.
G12	Clean install, type-check and integration execution	Unverified	Lockfile omits `xlsx` dependencies; prior `npm ci` failed, leaving `tsc`/`tsx` unavailable. No tests passed in this review. SRC18.

H. Expanded in-house business operations

These nine checks cover the latest product direction. Pharmacy-only helpers and recurring-medicine subscriptions do not prove a general business operations/SaaS implementation.

ID	Capability	Status	Verified scope and remaining work
I01	Standalone business and optional listing	Missing	No general business/module/publication settings identified. Internal operations must work with public listing off, retaining private patient and commercial data. SRC16/SRC19.

ID	Capability	Status	Verified scope and remaining work
I02	Direct/app/phone/import origin and referrals	**Partial**	Pharmacy online/in-store and in-person/assisted consultation foundations exist. Add consistent cross-business origin, independent referrer and commission attribution without rewriting old references. SRC06/SRC09/SRC19.
I03	Lab/clinic/hospital supplies inventory	**Missing**	Inventory models/services discovered are pharmacy/medicine-specific. Add non-medicine items/units, stock locations/lots and movement balances; preserve medicine safety rules. SRC04/SRC19.
I04	Department stores and issue/receive transfers	**Missing**	A generic transfer adjustment reason is not a complete branch/location dispatch-receipt/custody workflow. Add stores, permissions, in-transit balance and receiving reconciliation. SRC04/SRC19.
I05	Visit/test/procedure supply consumption	**Missing**	No dedicated per-service consumable usage/waste/reversal workflow identified. Post once against a test/visit/procedure; avoid duplicate consumption or unintended duplicate patient charge. SRC19.
I06	Shared private business billing and cash close	**Partial**	Fees, order totals, manual paid marking, wallets and specialist payout documents exist. Full cross-business invoices/credits/receipts/dues/cashier close remain incomplete. SRC06/SRC11/SRC12/SRC19.
I07	Private versus public tariff/catalogue	**Partial**	Retail/wholesale fields and provider catalogue overrides exist. Add controlled publication, private tariffs and immutable sold-price/source snapshots across business types. SRC04/SRC06/SRC10/SRC19.
I08	Business software subscriptions and entitlements	**Missing**	Existing subscriptions track recurring medicines. No general SaaS plan/module/billing-period entitlement implementation found; keep platform fees separate from customer bills/marketplace commission. SRC07/SRC19.
I09	Private business onboarding and opening balances	**Missing**	Pharmacy medicine imports provide a limited reusable pattern. No complete cross-business private patient/service/item, opening supply stock/dues import and reconciliation workflow found. SRC05/SRC15/SRC19.

Corrections to the first plan

1. **Low stock is already implemented as an API query.** Retain it; build alert delivery/replenishment operations only where needed.
2. **Prescription dosage/frequency/duration/instructions exist.** Add explicit strength/route, stronger finalization and amendment rules; do not recreate the fields.
3. **Connected counter order and manual cash/COD marking exist.** The remaining gap is a complete POS/register/invoice/reconciliation workflow.
4. **In-person mode and pharmacy-assisted consultation booking exist.** A reception/walk-in queue and safe shared capacity still need implementation.
5. **Specialist payout accounting and invoice/receipt assets exist.** Doctor self-service visibility, general vendor settlement and bank confirmation are distinct work.
6. **Refill reminder preferences are not scheduled dispatch.** The source registers booking-reminder and pharmacy-order side-effect consumers, but no due-refill dispatcher was identified.
7. **Stock-return helper presence is not an end-to-end sales return.** Batch identity, credit document, return authorization and refund reconciliation still need a designed workflow.
8. **Basic report upload/retrieval and API lab integrations exist.** The missing lab work includes the partner panel for labs without APIs, assigned order/sample status, signatory/amendments/review and secure event delivery. A full LIS remains optional.

Source evidence register

All paths below resolve in the reviewed checkout. Function/field names identify the evidence; the root planning docs are not used as proof of their own claims.

* **SRC01:** pharmacy model `(src1/modules/pharmacies/model/index.ts)`, pharmacy routes `(src1/modules/pharmacies/routes/index.ts)`, role guards `(src1/modules/middleware/auth.ts)`: active/verified pharmacy guards, document fields.
* **SRC02:** medicine model `(src1/modules/pharmacyMedicines/model/index.ts)`, medicine routes `(src1/modules/pharmacyMedicines/routes/index.ts)`: catalogue, prescription/tax metadata and discovery.
* **SRC03:** cart service `(src1/modules/pharmacyCart/services/index.ts)`, `checkout`: actor/buyer routing, refreshed prices, `allowPriceOverride: false`, self-purchase denial; cart routes `(src1/modules/pharmacyCart/routes/index.ts)`.
* **SRC04:** inventory types `(src1/modules/pharmacyInventory/types/index.ts)`, inventory repository `(src1/modules/pharmacyInventory/repository/index.ts)`, `reserveBatches`, `atomicReturn`; inventory routes `(src1/modules/pharmacyInventory/routes/index.ts)`: low-stock and adjustment endpoints.
* **SRC05:** import service `(src1/modules/pharmacyInventoryImports/services/index.ts)`, import routes `(src1/modules/pharmacyInventoryImports/routes/index.ts)`: preview/commit/retry/error download/cleanup.
* **SRC06:** order model `(src1/modules/pharmacyOrders/model/index.ts)`, order service `(src1/modules/pharmacyOrders/services/index.ts)`, `create`, `markPaid`, `resolveOrderItemsPricing`, `assertPrescriptionProvided`; order routes `(src1/modules/pharmacyOrders/routes/index.ts)`.
* **SRC07:** subscription model `(src1/modules/pharmacySubscriptions/model/index.ts)`, subscription service `(src1/modules/pharmacySubscriptions/services/index.ts)`, `reconcileForOrder`; registered consumers `(src1/modules/queue/service/consumers.ts)`: two consumer types, no identified refill dispatcher.
* **SRC08:** specialist routes `(src1/modules/specialists/routes/index.ts)`, portal service `(src1/modules/specialistPortal/services/index.ts)`, prescription validation `(src1/modules/specialistPortal/validation/index.ts)`: portal operations, current-object prescription updates and PDF generation.
* **SRC09:** booking types `(src1/modules/bookings/types/index.ts)`, `IBookingPrescriptionMedicine`; booking controller `(src1/modules/bookings/controller/index.ts)`, `resolveBookingUserId`; booking service `(src1/modules/bookings/services/index.ts)`, `createSpecialistBooking`; slot repository `(src1/modules/slots/repository/index.ts)`, `incrementBookingCount`.
* **SRC10:** external booking service `(src1/modules/externalBookings/services/index.ts)`, LabStack client `(src1/modules/bookings/services/labstack/index.ts)`, Healthians client `(src1/modules/bookings/services/healthians/index.ts)`, webhook routes `(src1/modules/webhooks/routes/index.ts)`, webhook service `(src1/modules/webhooks/services/index.ts)`.
* **SRC11:** payment service `(src1/modules/payment/services/index.ts)`, `verifyAndMarkPaid`, context processing and wallet refund methods; user wallet `(src1/modules/wallet/services/index.ts)`, pharmacy wallet `(src1/modules/pharmacyWallet/services/index.ts)`.
* **SRC12:** payout routes `(src1/modules/payouts/routes/index.ts)`, admin guard; payout service `(src1/modules/payouts/services/index.ts)`, `createDisbursal`, `getInvoice`, receipt generation; payout types `(src1/modules/payouts/types/index.ts)`.
* **SRC13:** file routes `(src1/modules/file/routes/index.ts)`, file controller `(src1/modules/file/controller/index.ts)`, `serveFileByPath`; central routes `(src1/routes/index.ts)`: public static mount.
* **SRC14:** booking reminder scheduler `(src1/modules/bookings/services/bookingReminderScheduler.ts)`, task queue `(src1/modules/queue/service/TaskQueue.ts)`, job consumer `(src1/modules/queue/service/JobConsumer.ts)`, notification service `(src1/modules/notification/services/index.ts)`.
* **SRC15:** user model `(src1/modules/user/model/index.ts)`, dependent model `(src1/modules/dependents/model/index.ts)`, pharmacy customer service `(src1/modules/pharmacyCustomers/services/index.ts)`, patient tracking

`(src1/modules/patientTracking/services/index.ts)`, partner model `(src1/modules/partners/model/index.ts)`, reward referral service `(src1/modules/referral/services/index.ts)`.

* **SRC16:** Complete module/schema/route inventory plus code-only graph searches for patient/consent/encounter/organization/purchasing/invoice/specimen/hospital/care-task symbols. Targeted fallback code searches found no `specimen`, `accession`, `clinicalReferral`, `consentVersion`, `cashRegister`, `purchaseOrder`, `goodsReceived`, `creditLimit`, `invoiceNumber`, `tenantId`, `organizationId`, `wardId` or `bedId` implementations. Absence claims are limited to these workflows and this checkout, and account for unrelated OAuth/reward-referral symbols.
* **SRC17:** OTP service `(src1/modules/otp/services/index.ts)`, configuration `(src1/config/config.ts)`, user controller `(src1/modules/user/controller/index.ts)`, Meet routes `(src1/modules/meet/routes/index.ts)`, central routes `(src1/routes/index.ts)`. Credential findings were validated by file/key-presence metadata without copying secret values.
* **SRC18:** package manifest `(package.json)`, lockfile `(package-lock.json)`, legacy entry `(server.js)`, maintained entry `(src1/index.ts)`, pharmacy integration cases `(test/pharmacy.integration.test.ts)`. See the prior exact failed-command results in repository review.
* **SRC19:** Expanded-scope graph search found inventory/import classes under pharmacy modules and subscription classes under `pharmacySubscriptions`, with no dedicated supplies/warehouse/stock-consumption/software-plan classes. Inventory model `(src1/modules/pharmacyInventory/model/index.ts)` and types `(src1/modules/pharmacyInventory/types/index.ts)` use pharmacy/medicine references; subscription model `(src1/modules/pharmacySubscriptions/model/index.ts)` uses medicine/refill fields. Generic lab/clinic/hospital business workflows are requirements, not inferred implementations. The complete module inventory remains in repository review.

Plan impact

The expanded scope adds private business/module/publication controls, supplies inventory/consumption, business billing and SaaS plan records. M2/M3 deliver clinic/pharmacy and shared supply foundations; M4 adds direct lab operations plus API/panel routes; M5 adds hospital stores and broader business workspaces. M0 must be re-estimated; T0 is its accepted completion. M1 weeks 1-3 are provisional and require a finance-cutover-inclusive re-estimate; rebase later windows if needed. F14 phone access is separately estimated after M1/M2; any M4 pilot is conditional on staffing and its funding/clinical/telecom gates. Indicative pilot post-M0 weeks 11-12 and broader rollout weeks 19-24 replace project-start calendar ranges, subject to the same team and release gates. Existing source is still reuse-and-verify scope; no application features were added by this documentation change.

No arbitrary percentage-complete score is assigned. Runtime correctness, frontend completeness, real-provider behavior and the 100 RPS claim remain unverified. The performance plan defines how to establish that evidence.

2. Daily business operations and optional marketplace

Source: documentation/planning/BUSINESS_OPERATIONS.md

Product direction: support pharmacies, laboratories, doctors/clinics and hospitals as daily business-management customers, including sales and services received outside the Jansetu app. **Planning update:** 8 November 2026. **Code baseline:** `9ca8b54606581f2158e760cad79aa7b845529184`; the capabilities below are requirements, not new implementations.

The platform has two connected uses: private business operations and the patient-facing marketplace. A business can start with its internal workspace and later enable approved marketplace services. The same model supports Jansetu-operated facilities and independent partners; ownership is recorded separately from the services a business provides. See the roadmap and current capability evidence.

1. Scope decisions and working rules

Decision	Planning rule
Daily operations	Each business can manage its own patients/customers, walk-ins, appointments/orders, invoices, receipts, purchases, stock, returns and staff within enabled modules. A Jansetu-generated order is not required.
Optional public listing	Internal operations do not require a public marketplace listing. Publication is explicitly enabled for approved branches/services; disabling discovery does not discard accepted bookings or private records.
Business types	A business may operate a pharmacy, lab, clinic, hospital or several of these. A hospital's pharmacy/lab can be departments under that business; independent suppliers retain their own business ownership.
Channels and attribution	Record the primary origin channel: marketplace app, walk-in, phone or approved import/API. Record a referring person/business independently, since a referral may arrive through any channel. An online payment does not automatically make a walk-in a marketplace sale.
Operations versus integration	Marketplace participation and API availability are separate choices. A listed lab can use its APIs or the panel; an unlisted lab can use the panel for its own direct customers.
Stock authority	One authoritative owner for each business/branch/item/location balance. App, counter and procedure consumption affect that owner's stock; an external ERP-owned balance remains an integration/read projection unless authority is deliberately transferred.
Patient identity and privacy	A patient can have one permanent Jansetu identity plus business-local references. An internal customer does not need an app login or public profile. Record linkage does not grant another business access to clinical history.
Basic-phone access	Approved pharmacies/labs/hospitals can identify a patient and fund a consultation account under the selected arrangement; a basic-phone helpline reaches doctors without an app. Patient identity, partner funding, care consent and clinical access remain separate. See PHONE_CONSULTATIONS.md.
Adoption and continuity	Make daily workflows useful through onboarding/imports, fast counters, connected records, worklists and reports. Include authorized export, backup and recovery as normal capabilities.
Commercial model	Proposed: business software subscription plus separately configured commission on marketplace-attributed transactions. Rates, plans, collection responsibility and settlement contracts remain decisions to approve; do not assume commission on direct business.

The optional-marketplace rule is the planning default adopted from the preceding discussion. Commercial prices, feature packages and final invoicing entities remain open decisions.

2. Workspace requirements by business

Workspace	Daily services and sales	Inventory and purchasing	Financial and operations views
Pharmacy	Walk-in/app sales, retail/wholesale entitlement, prescription review, dispensing, pickup/delivery and returns	Medicines, batches, expiry/recall/quarantine, UOM, suppliers, purchase/receive, reservation, transfer and approved return/restock	Sales invoice/credit note, cash/online receipts, shift close, supplier balances, margin and stock movement
Laboratory	Direct/walk-in/referral/app test orders, appointment/collection worklist, sample status, original/corrected reports, signatory release and clinician review	Reagents, kits, vials and consumables; purchase/receive, units, lots/expiry where applicable, consumption, wastage and stock count	Test invoice, receipts/refund, receivables, referral/marketplace attribution, turnaround backlog and partner statement
Doctor / clinic	Own appointments, walk-ins and app visits, clinical records, prescriptions, procedures, fees and follow-up	Clinic consumables and supplies; receipt, use against visit/procedure, wastage, stock count and low-stock worklist. Medication dispensing uses the pharmacy-specific workflow when enabled	Visit/procedure invoice, receipts, day-close, pending dues and separate marketplace earnings/settlements
Hospital	Registration, reception/OPD queue, consultations, procedures, diagnostics/pharmacy coordination and service billing	Central/department stores, receipt, issue/receive transfers, consumption, count/adjustment, lot/expiry where applicable; medication stock retains pharmacy rules	Department tariff/invoice, receipts/refunds, cashier close, purchases/payables, department usage and OPD operations. IPD/nursing/OT remain later scope

Basic lab/clinic/hospital inventory is now initial business scope. It must not be deferred under the label of optional full LIS or future IPD. Analyzer interfaces, QC/calibration, instrument-generated consumption, full general ledger accounting, insurance/claims and advanced asset maintenance remain separate extensions unless specifically selected.

Typical daily journeys

* **Pharmacy:** receive a supplier batch, sell at the counter and through the app, reconcile the same stock, then close cash and process an authorized return.
* **Lab:** register a direct patient without an app account, book a test, collect a sample, post approved supply consumption, receive payment and release a verified report. An app-origin order uses the same lab worklist and record rules.
* **Clinic:** add a walk-in, record a visit/procedure, issue applicable supplies, take a payment, print/send an authorized receipt and schedule follow-up.
* **Hospital:** register an OPD visit, issue supplies from a department store, collect the service charge, receive an internal stock transfer and reconcile the cashier's day.

3. Shared product and data boundaries

Concept	Required meaning and rules
Business, branch and department	Scope staff, finances, catalogue and records by trusted membership. A branch can have multiple stock locations or departments without becoming an unrelated business.
Enabled modules	Configure clinic, lab, pharmacy, hospital and inventory capabilities explicitly. Do not assume every business needs all modules or build one identical screen for every role.
Private and public catalogue	Keep internal service/items and private tariffs separate from approved marketplace publication. An internal price or stock quantity does not become public by enabling a staff workspace.
Order/visit origin	Preserve source, business, branch, creator, external reference and referring party. Keep commission attribution separate from price class and payment method. Retain legacy online/in-store fields through compatibility mapping.
Item and service	A billable test/consultation is a service. A vial, glove or reagent is an inventory item. A service can consume items without selling each item separately to the patient.
Inventory location and lot	Identify item, stock unit, business/branch/location, optional lot/batch/expiry and quality status. Reagents may use fractional units such as mL; define conversion, decimal precision and rounding explicitly instead of inheriting integer medicine quantities. Medicine prescription/allocation rules remain in the pharmacy boundary.
Stock movement	Append attributable receive, reserve, release, consume, sale, return, transfer, waste, count adjustment and reversal records. Balances must reconcile to movements; edits cannot silently erase previously accepted stock effects.
Service consumption	Link a consumption record to its visit/test/procedure and authorized actor. Start with explicit quantities or an approved configurable usage list; allow documented differences and waste. Replayed status/report events cannot consume stock twice.
Billing and collection	Distinguish service/product invoice, credit note, cash/gateway receipt, outstanding balance and refund. A completed clinical service or report release is not itself evidence of payment.
Business and platform finance	Separate the business's customer/supplier balances from Jansetu's subscription invoices and marketplace commission/settlement records. Cash collected by a business is not a receipt into Jansetu's bank account.
Software subscription	Track plan, enabled modules/limits, billing period and subscription state separately from medicine-refill subscriptions. Start with reviewed plan assignment and billing records; automated recurring collections can be a separate integration.

Architecture extension

Retain the existing pharmacy services, FEFO transactions and safety rules. Introduce a focused non-medicine supplies inventory boundary for lab/clinic/hospital stock and a shared purchasing/billing contract. Share movement/idempotency primitives only where doing so preserves domain rules; do not replace the pharmacy engine with a generic item table before parity tests pass.

`organizations` owns business/branch/staff membership and module/publication settings. `billing` owns business invoices/receipts/credits. Proposed `suppliesInventory` owns non-medicine item/location/lot/movement and consumption. `procurement` owns supplier/purchase/receive references. Existing pharmacy modules retain medicine stock authority. Proposed `businessSubscriptions` owns software-plan records; existing `pharmacySubscriptions` continues to mean recurring medicines.

One order/visit can reference one commercial invoice, payment records, authorized stock movements and clinical documents without merging those states. Each aggregate has one authoritative writer. Integration adapters may read projections or issue commands to that writer rather than independently adjusting balances.

4. Laboratory API and panel rules under the expanded scope

1. Keep **API-connected third-party labs** and **panel-managed third-party labs**. Lack of an API does not prevent participation or force a complete LIS installation.
2. The panel now handles the lab's direct/walk-in/referral work as well as Jansetu assignments. Add lab-local registration, service tariffs, invoices/receipts and consumables inventory to the earlier order/report panel scope.
3. Record origin separately from `fulfillmentMode: api` or `panel`. A direct panel order must not automatically create a duplicate order at an API provider. An app order routed through an API must retain its provider reference.
4. Choose the owner of each stock/result/order resource. If the lab's existing system owns reagents, use a permitted integration projection; if Jansetu owns supplies stock, post its movements only in Jansetu. A provider callback cannot independently decrement the same balance twice.
5. Both routes preserve report verification, original/amended versions, clinician review, critical-result acknowledgment and the same durable release events. Internal lab use does not grant unrestricted access to another business's patient records.
6. Manual or configured supply consumption is in scope. Analyzer connectivity, automatic instrument consumption, QC and calibration stay optional later integrations.

Basic-phone consultations and counter top-ups

The phone-channel specification defines registration and patient/PIN verification, funded partner top-ups with SMS/printed receipts, language selection, disclosed fixed-fee reservation, doctor bridge/callback, no-answer release and dropped-call reconnect. Shared phones do not imply shared balances/charts; no-phone patients retain assisted/caregiver access. Reception/cashier permission does not confer clinical access or PIN reset. Partner cash is reconciled separately from platform-held funds; a top-up is spendable only through the approved funding contract. This is F14, separately estimated after accepted M1/M2 foundations and finance/legal/clinical/telecom decisions, with a conditional staffed M4 pilot. P1 chooses Partner-paid consultation, Consultation-only balance or General patient wallet before F14 sizing. Existing unrouted pharmacy sponsorship is qualified reuse with a paid-state guard; pricing, validation and durable idempotency remain prerequisites. PIN provider payloads are transient, wallet debits exclude held funds, and clinical UI/backend enforce mode/type prescribing with disclosed referral-only fee policy. Private patient/caregiver PIN setup and bounded recovery, independent contact proof, staged abuse continuity and same-practitioner/prior in-person follow-up evidence are explicit F14 gates. The backlog retains every original top-up/receipt, queue/capacity, one-charge/reconnect and authorized outcome deliverable alongside these controls. No new M0 dependency or automatic ABHA prerequisite is added.

5. Commercial, access and lifecycle decisions

* Keep subscription charges for using the software distinguishable from marketplace commissions. Proposed default: direct orders have no marketplace commission; a business software subscription may still apply. Final rates and eligibility need a commercial decision.
* Record who issued the customer invoice, collected the payment and owes the refund. Preserve financial-period numbering, original invoices and linked reversals. Finance must approve the configured seller/invoice entities and tax treatment before release.
* API capability does not determine the business's subscription or public listing. A business may buy internal modules and selectively publish services later.
* Revoking marketplace listing stops new public discovery/bookings according to policy; accepted orders, refunds, support cases and authorized record access still require completion. A subscription state change cannot silently delete clinical or financial history.
* Require individual staff accounts, role/branch/location scope, authorized exports, joiner/leaver controls and reasons for price/stock/financial overrides. Reception, cashier and inventory access do not automatically grant clinical editing rights.

* Import private patients, item/service masters, opening balances and receivables using reviewable templates and deterministic import keys. Never assume phone equality authorizes a patient merge, or invent clinical consent, lot/expiry or payment evidence during import.

Architecture-review clarification: preserve existing platform-configured gateway collection and wallet-credit evidence during migration; record customer wallet liability and platform-business dues, rather than labelling a wallet credit as an original-method refund. M0 applies atomic wallet effects and factual classification without choosing new refund destinations; finance approves the future account map and reconciled opening balances before M2 business posting. Lab reports preserve preliminary/final/unknown completeness and immutable corrected versions; M0 fixes current gateway recovery and report overwrite before later suite expansion. Counter price overrides remain authorized, limited and actor/reason-audited with server-recomputed totals.

6. Scope, sequence and acceptance

M0 duration is **re-estimate at kickoff**. T0 is accepted M0 completion; subsequent windows below are relative to T0, not project start. The limited pilot is tentatively post-M0 weeks 11-12 and expanded rollout weeks 19-24, using the same team and dependency assumptions. M1 must include financial classification, review and cutover in its estimate; rebase downstream windows if it extends. F14 phone work needs a separate staffed estimate and is not included in the stated windows. Earlier project-start estimates are superseded; recalculate dates after M0, shared contracts and the first import rehearsal.

Stage	Expanded deliverable
M0, re-estimate at kickoff; ends T0	Build/security/access, current-flow payment recovery, immutable-report bridge, create-only correctness-index/readiness gates and safe staging
M1, post-M0 weeks 1-3 provisional; re-estimate including financial cutover	Business/branch/membership, standalone/publication settings, patient identity/consent and common billing/item/location contracts
M2, post-M0 weeks 4-6	Doctor/clinic appointments, clinical workflow, direct/app attribution, fees, business invoices and receipt foundations
M3, post-M0 weeks 4-8	Pharmacy/POS plus supplies inventory/purchasing, location balances, manual consumption, returns and cashier close; integrates M2 billing
M4, post-M0 weeks 9-12	API/panel lab routes plus direct lab orders, billing, consumables, report release and care continuity; gated pilot
M5, post-M0 weeks 13-18	Hospital OPD/department stores, broader vendor workspace, subscription-plan/billing records, private/public tariffs and business reporting
M6, post-M0 weeks 19-24	Cross-business regression, migration/restore, useful 100 RPS mixed load and staged rollout

F14 phone pilot follows M1/M2 and P1-P3 acceptance; joining M4 is conditional on staffing/re-estimation. The separate phone specification owns the full acceptance list.

Before each business type goes live, demonstrate:

1. An unlisted business registers a direct customer and completes an in-house service/sale without public app listing or patient app login.
2. Public discovery and other businesses' searches do not expose that business's private patients, prices, stock, invoices or reports.
3. App and direct orders use the correct business/branch, tariff and source. A refund, referral or payment retry does not rewrite the original attribution or duplicate commission.

4. Simultaneous sale/consumption/transfer commands cannot overdraw stock; warehouse transfer dispatch/receipt and stock-count adjustments reconcile to the movement history.
5. A lab test or clinic procedure posts approved consumables once, including documented waste/reversal. Consumables included in a service price are not billed again unintentionally.
6. Sales, receipts, credits, outstanding balances, supplier payables and cash close reconcile. Subscription bills and marketplace settlements remain separately identifiable.
7. A business can enable or disable eligible public services while completing existing orders and preserving authorized records.
8. Imports/replays preserve source IDs, opening quantities, lot/expiry provenance and financial totals. Staff revocation, export, backup and restore behave as documented.

Use the performance plan for a workload that includes private business operations as well as marketplace requests. No software implementation, subscription charging, partner onboarding or production migration is performed by this documentation update.

Phone contact authority (AR-25): a counter-entered number stays unverified until handset possession and independent patient/caregiver binding are approved under M1 identity policy. Cashiers cannot self-approve contacts, set/recover PINs or obtain protected verification priority; legitimate shared phones remain supported. The disposable setup slip shows the masked contact for readback. Follow phone3.4, including staff-number/unrelated-patient reuse review and separate key-lifecycle approvals in P3.

v1.7 current-path and restore contracts: shared catalogue authority is separate from local stock/offer permission; quantity commands retain quality holds, lot-based applicable price ceilings and confirmed quote versions govern sales, and appointments bind the correct practitioner/resource interval. Restored stale authority/stock/money is held behind D9 admission and per-effect reconciliation. D1/D4 precede M1 billing/settlement contract freeze, then M2 posting. See principal review disposition `(documentation/planning/PRINCIPAL_REVIEW_DISPOSITION.md)` and M0 approval scope `(documentation/planning/M0_IMPLEMENTATION_APPROVAL.md)`.

3. Basic-phone consultations and assisted top-ups

Source: documentation/planning/PHONE_CONSULTATIONS.md

Version 1.7, 9 November 2026. Status: user-requested scope; proposed design, not implemented. A patient without a smartphone can fund a consultation account at an approved pharmacy, laboratory or hospital, then call a Jansetu helpline from a basic phone to speak to an available doctor. This is a top-up of a healthcare account, not mobile-network airtime. No ABHA, app login or smartphone is required for this route. AR-23/AR-24 restore cumulative F14 scope and add private PIN enrolment/recovery, staged abuse response and explicit follow-up evidence alongside earlier controls; AR-25 defines contact verification and independent patient/caregiver binding; the supplied positive verdict concerns v1.5 and is not independent approval of this revision.

The system design `(documentation/planning/SYSTEM_DESIGN.md)`, roadmap and business operations remain the shared contracts. This document adds the channel without creating a second patient, wallet, appointment or clinical-record authority. Fixed consultation fees, callback behavior and charge criteria below are proposed defaults for owner review, not invented commercial terms.

1. Patient and partner journey

1. **Register or identify the patient.** Authorized staff creates/finds the permanent Jansetu patient ID with a phone contact or authorized caregiver. A family phone can serve several patients; the caller selects the patient. The phone is a contact/lookup hint, never the unique patient identity or spending authority. Staff-entered contacts start unverified; section3.4 defines confirmation and independent binding before credential use. Patients with no phone retain the assisted reception route and can receive a printed account reference.
2. **Confirm and fund the right account.** Staff confirms a minimally disclosed patient identity and amount, records the patient beneficiary separately from payer/caregiver, obtains approval, and issues an SMS or printed receipt with a transaction reference. Partner top-up permission alone cannot reveal the patient's history, change their PIN or spend their balance. First-use PIN setup happens privately through section3.3; no cashier terminal or receipt contains the patient's PIN.
3. **Call or request callback.** The patient calls a published Jansetu number, chooses a supported language and enters patient reference plus a private PIN, or uses an approved assisted-verification route. Caller ID or possession of a shared phone is insufficient. SMS is optional: keypad prompts, voice assistance and printed references must work without it. Callback support and toll-free versus normal call charges must be disclosed; an ordinary incoming number may still require the patient's mobile-network airtime. Clinical operations must provide urgent-help/referral routing before a payment gate; urgent requests cannot wait behind a recharge or ordinary consultation queue.
4. **Agree to the consultation.** Establish the patient/caregiver authority, consent and appropriate doctor/service. Announce the fixed service fee, any disclosed access charge and the policy for advice/referral-only outcomes and refunds, check availability, then atomically reserve funds and doctor capacity. Do not promise a prescription. Offer a bounded queue or consented callback when unavailable. Waiting, unanswered and failed-to-connect calls do not consume the consultation fee.
5. **Connect to a doctor.** Bridge the patient and verified practitioner through a telephony provider. Use the same appointment/encounter as an app consultation and show the authorized clinical context in the doctor dashboard. A telecom answer event is not evidence that a clinical consultation occurred. The clinician determines whether audio is adequate or a video/in-person review is needed. Offer a returning patient their previous practitioner when available and consented, or an explicitly explained wait/callback; another available doctor does not inherit follow-up prescribing eligibility.
6. **Finish or reconnect.** Under the approved charge policy, post one fee against the same consultation operation when the clinical service becomes chargeable. A dropped call resumes that operation during a disclosed reconnect window; it does not automatically create another fee. Uncertain outcomes enter reconciliation/manual review. Avoid per-minute consultation debits and automatic clinical-call termination when a running balance reaches zero.
7. **Deliver the outcome.** Provide a minimal receipt/status by the chosen channel and authorized prescription access/printing. Avoid sensitive details in shared-phone SMS or spoken prompts. A pharmacy/lab/hospital receives

clinical documents only under the relevant patient authority and release policy. Recharge/referral attribution is separate from care consent and does not automatically create commission.

2. Top-up funding, balance and reversals

Customer balance liability, partner funds, partner cash collections and doctor/business payables are distinct. Recording cash at a counter is not proof Jansetu received money. Proposed first-pilot funding is a prefunded partner allocation or confirmed digital collection under an approved funds-flow contract. If a receivable/credit facility is later selected, it needs explicit limits and settlement terms; there is no implicit unlimited partner credit.

Funding choice and existing reuse (AR-21)

P1 must select the model before F14 is sized; fund custody, settlement and operating approval still gate any live use. The requested reloadable patient balance remains in scope. The Partner-paid consultation option is a narrower route, not a replacement silently selected by this document.

Model	Account and care boundary	Reuse / unresolved work
Partner-paid consultation	Debit partner funds for one identified patient's consultation; no unspent patient credit is created by that operation	Existing pharmacy service is a primitive. Partner funds, collection and doctor settlement still need review; no automatic RBI exemption is claimed
Consultation-only balance	Reloadable patient credit restricted to approved consultation charges; a purpose-restricted pocket under the shared wallet authority, with its own available/held counters	Preserves the requested top-up journey without authorizing other marketplace spending; identity, funding, expiry/reversal and access rules need P1 approval
General patient wallet	Credit the approved account holder's general balance; each admitted use and payer/beneficiary relationship is defined explicitly	Existing customer wallet code is reuse material, not approval for universal third-party spending; scope, migration and financial obligations require P1/D1/D2 review

`PharmacyWalletService.payConsultation` already checks a linked pharmacy/patient and transactionally debits the pharmacy wallet, records a movement and conditionally marks the booking paid. The paid-state guard can roll back a competing debit; saying it has no duplicate protection would be inaccurate. However, it accepts `req.amount`, has no dedicated payment-body schema or durable command/result idempotency, and the inspected pharmacy-wallet router exposes only balance/history/top-up. This is unrouted, unverified pharmacy-only reuse, not a live phone/payment feature or a lab/hospital wallet. It uses app-account-linked bookings, so M1 identity work remains necessary. Service `(src1/modules/pharmacyWallet/services/index.ts)`, router `(src1/modules/pharmacyWallet/routes/index.ts)`, validation `(src1/modules/pharmacyWallet/validation/index.ts)`.

Before any exposure, derive the payable fee/currency and remaining obligation on the server from authorized booking/price evidence; validate allowed inputs, recheck ownership/eligibility and expected state in the transaction, and return the original result for an authenticated duplicate command. Retain the existing paid-state guard. Do not enable a route, accept a caller-selected fee or generalize pharmacy funding to labs/hospitals through this documentation update. All chosen models use the same financial owner; a new pocket cannot become an independent balance writer.

Each top-up has a stable economic-operation ID, patient beneficiary, payer, currency/unit, collecting business/branch/staff, funding source, receipt and status. In one transaction, a prefunded top-up conditionally reduces that partner's available allocation and credits the patient liability with matching movements and outbox/result. Gateway funding credits only after captured/settled evidence according to the selected model. Duplicate clicks, callbacks and crashes cannot mint additional value. Declined or unknown funding cannot appear as spendable balance.

Resolve the beneficiary to the approved account holder/payer through the M1 identity crosswalk and caregiver authority. Existing `userId` wallets cannot simply be relabelled with a phone or patient ID; account ownership and any legacy balance mapping need reviewed evidence.

Consultation funds have available/reserved/spent/released states. Available balance excludes active reservations; competing app and phone spends use guarded updates on the same account. Repeated calls for an active consultation resolve that operation rather than reserve twice. Reservation expiry must reconcile whether a bridge or consultation is active; a delayed telecom callback cannot revive a released reservation or charge a new session. Active clinical sessions use bounded leases and an operator escalation path, not silent expiry or an indefinite money hold.

The authoritative account/pocket keeps posted and held minor-unit counters with `available = posted - held` and `posted >= held >= 0`. Reserve guards available and increases held; release decreases held once; capture of that hold decreases posted and held once. Ordinary spend, transfer and other reservations guard available on the same document, not a sum of hold rows read from a snapshot. The operation, hold/movement, booking state and outbox/result commit together. App spending cannot consume phone-held funds, and a restricted pocket cannot fund a disallowed service.

Unspent top-up reversals and refunds preserve the original funding/beneficiary references. Partner removal cannot erase customer liability. Refunds to wallet, gateway or cash share the captured-payment refund budget described in design section 9.4; unknown external refunds keep their reservation until reconciled. Do not authorize a cash-out, transfer or new refund destination through this feature before D1/D2 and P1 below are resolved.

3. Architecture and provider contracts

`phoneConsultations` is a proposed application module in the existing modular backend. It coordinates patients/consents, booking capacity, billing/wallet operations and clinical encounter commands through their owners. A telephony port provides number/call bridging, callback/status lookup and termination controls; its adapter owns provider payloads, not patient balances or clinical release. An IVR is a voice menu reached through the telephone network; the patient does not need mobile data. A partner panel still needs the connected counter service.

Keep one consultation ID and stable provider call-leg mappings. Persist an outbound call intent before provider submission, and do not redial blindly after an unknown timeout. Signed/authenticated ordinary status events enter the durable inbox; replay and out-of-order handling are version/fence guarded. PIN-bearing input follows the transient verification exception below and never enters a raw event archive. Workers reconcile provider facts outside Mongo transactions. Provider-secret rotation, restricted egress, retention and redacted logs apply. Call lifecycle and charge lifecycle are separate: ringing/answered/disconnected cannot directly declare a completed encounter or debit the wallet.

Persist call/consent/charge metadata with restricted access. Do not record calls by default. Recording, if specifically selected, requires its own consent, purpose, access and retention decision; it is not a prerequisite for keeping the clinician's encounter notes. A top-up cashier cannot reset a patient PIN; recovery uses separately authorized, audited identity checks, including changed/recycled numbers.

3.1 PIN input, verification and abuse controls (AR-20)

Prefer an authenticated provider POST with keypad input in its body; do not accept PINs in URL/query/path. Verify the provider signature over bounded raw bytes in memory, verify the call/session binding, apply shared admission limits before expensive verification, check the PIN transiently and discard the digits. Persist only an allowlisted event/session reference and non-secret processed outcome. No raw PIN payload, keypad digits or replayable verification grant enters the inbox/outbox, dead letters, audit, traces, request dumps, exception bodies or recordings. Disable sensitive request capture at the edge, proxy, framework, APM and provider console; a plaintext URL can leak before application redaction. Provider retention/DTMF behavior is unverified until P3 acceptance. Reject unsupported transports/configurations rather than archive the input. OWASP logging guidance.

Use a vetted slow password KDF with unique salt and an additional secret pepper/key in a separately access-controlled vault/HSM/KMS-backed secret, outside MongoDB and its backups. Store only the approved keyed verifier and key version, not a separate unkeyed verifier or plaintext PIN. Define key outage, rotation and compromise/reset procedures before release; hashing alone does not solve a short PIN's small search space. A database-only theft test must not permit local PIN verification without the separate key; this does not claim safety after key/service compromise. OWASP password storage guidance.

Issue random non-sequential references with check digits and enough entropy for the expected population, tested for keypad usability. A check digit detects input errors and is not authentication. Deny common/repeated/sequential PIN choices. Enforce shared, multi-instance attempt/call budgets per patient/account, session, calling number/risk signal, ingress/provider and globally; caller ID is not a trusted identity. Generic failures and minimal prompts limit account enumeration. A one-guess-per-account attack must trigger the global abuse response, not evade it. Apply admission limits before expensive verification. A global signal must not simply disable PIN checks for every patient: progressively slow/challenge risky attempts, limit abusive sessions/provider sources and reserve bounded verification capacity for established sessions or independently verified-contact callback. A protected lane still requires authentication, consent and spending authority; caller ID alone cannot qualify it. Existing authorized encounters and urgent-help/controlled assisted routes remain available within engineered capacity. KMS/key outages still fail closed for PIN verification;

throttling is not permission to skip the verifier. Any staff-assisted fallback is separately authorized, audited and throttled; it cannot bypass the account's spending/consent authority or give a cashier PIN-reset rights.

Successful verification gives a short-lived patient/purpose/consultation-bound authorization through the backend's identity owner, not an unscoped wallet-spend flag. Deduplicate provider retries by safe event/session identity without retaining the submitted digits. Repeated input must not mint multiple grants or reset attempt limits. Synthetic end-to-end checks must cover provider request/retention settings as well as all local persistence and observability surfaces; no provider has yet been certified to meet this contract.

The generic inbox payload-hash rule does not apply to PIN-bearing bytes. An unkeyed digest of short keypad input can itself enable guessing; retain no such fingerprint. Any retained conflict/receipt hash covers the allowlisted sanitized envelope only, with attempt identity and verifier state controlling replay.

3.2 Prescribing and referral-only outcomes (AR-22)

The cited 2020 guidelines distinguish medicine lists by actual consultation mode/type: List O covers any mode; first-consult List A requires video; eligible follow-up refills/add-ons have defined A/B scope; Schedule X and NDPS-listed substances are prohibited remotely. This is a policy baseline, not a medication catalogue or an assertion of current clinical certification. Guidelines sections3.6-3.7.4 and matrix.

Before release, P2 approves a versioned policy/catalogue against then-applicable rules and separately assesses audio suitability. Store actual audio/video mode, first versus eligible follow-up, current and prior practitioner IDs, prior encounter ID/date/mode, condition and policy version. The cited baseline requires the same practitioner within six months of a prior in-person consultation for the same condition, without disqualifying new symptoms or lost clinical context. Missing/changed practitioner, non-in-person or missing prior evidence, expired window or a different condition prevents automatic follow-up classification. Clinical owners approve date-boundary rules; do not silently equate six calendar months to 180 days. NRCeS guideline rendering. The doctor UI explains disallowed choices; backend checks block prohibited or mode/type-ineligible prescriptions even if a caller changes fields. Unknown classification or inadequate assessment routes to clinical review/referral, not a permissive default. A doctor using a web dashboard does not make a basic-phone encounter video, and a user-selected follow-up label cannot expand prescribing rights. Explain advice/referral-only outcomes and their fee/refund treatment before reservation; the clinician can choose referral without a prescription.

3.3 Private first-use PIN and recovery (AR-23)

Registration and funding do not activate a spend-capable credential. The patient sets the first PIN privately during their own IVR call or a backend callback to an independently verified patient/caregiver contact under section3.4. Both first-use paths enforce that binding before issuing a usable challenge or accepting setup. A caregiver may do so only under recorded authority for that patient and the relevant financial purpose. Partner accounts cannot set, read, choose, reset or recover a patient PIN, and no PIN is typed on a cashier terminal or printed on a receipt.

The service can issue a random, short-lived, single-use enrolment challenge bound to patient, verified contact, purpose and credential/contact version. If printed, use a separate disposable enrolment slip excluded from archived financial receipts/PDFs, panel history and logs. A cashier might see that slip: its code alone is insufficient. Also prove control of the independently verified contact, for example with a service-originated callback; caller ID is not that proof. The collecting partner cannot change that contact or reissue a usable challenge to a new contact without separately authorized identity checks. A missing independent contact uses the recorded caregiver/assisted identity route rather than an unverified PIN grant.

Only a protected one-time verifier and safe status/reference are stored. Enrolment/reset digits follow the same transient provider-input and no-recording/no-payload-fingerprint rules as PINs. Atomic expiry/version/unused checks consume the challenge once with the new keyed PIN verifier and credential version; concurrent/replayed attempts cannot activate twice. Rate-limit enrolment requests, challenge issuance, setup attempts, reset requests, reset attempts and contact changes by account, contact/session/source and staged global budgets. No unlimited recovery fallback exists.

Reset requires a new account-bound challenge plus verified-contact/approved identity evidence and any caregiver authority. Public demographic answers or a cashier's assertion alone cannot recover a credential. Merely requesting recovery must not change the PIN, invalidate valid sessions or lock out the victim. On completed reset revoke affected grants and older challenges, append secret-free audit and notify safe registered contacts by consented voice/SMS where available; notifications never include the PIN/challenge. Failed notification is retried and visible to support. Changed/lost contacts require a separate audited recovery decision, with notices to old/new safe contacts where possible. Preserve urgent-help access without granting unverified wallet authority. OWASP recovery guidance.

Before release, P3/security owners approve thresholds, time windows, protected-lane capacity and operator queues. Test that a legitimate verified-contact caller with the correct PIN completes verification during a defined cross-account spray below tested capacity. This is not an unlimited availability promise or a correct-PIN bypass of risk controls. OWASP authentication/lockout guidance.

No telephony provider, monthly minimum, per-leg cost, caller-number category or concurrent-call capacity is selected yet. Trial the provider's sandbox, event authentication, idempotency/lookup, language/keypad support, emergency/recovery behavior and total call-leg costs before committing the integration. Publish the support number only after end-to-end acceptance.

3.4 Contact verification and patient binding (AR-25)

A number entered by partner staff is **unverified**, including a legacy contact or a number with a generic OTP success. The identity owner records two separate facts: possession of the proposed contact and its approved binding to the patient or authorized caregiver. A successful call proves possession, not the beneficiary's identity. Neither caller ID, demographic answers, masked-number readback nor a cashier's assertion establishes that binding.

The service may place a rate-limited **contact-verification call** while the patient is present, with confirmation privately on the patient/caregiver-controlled handset. This unprivileged call contains no PIN setup code, clinical history or spending grant. It is distinct from a privileged **PIN-setup callback**, which an unverified contact cannot receive. Possession confirmation alone cannot issue an enrolment challenge, recover a PIN or admit a caller to the protected verification lane.

Before promotion to **independently verified**, require patient/caregiver confirmation and independently approved beneficiary/relationship evidence under the M1 identity policy. Use an already trusted patient credential or a separately authorized identity verifier checking approved evidence and caregiver scope; the collecting/registering cashier cannot approve their own submission by changing roles. A cashier answering their substituted number still fails this independent check. M1 identity/product/privacy and security owners must approve the evidence methods and assisted process; no ABHA, Aadhaar, smartphone or SMS is mandatory. If evidence is insufficient, retain a pending state and offer the audited caregiver/assisted identity route; urgent-help access continues without unverified wallet authority.

The identity owner alone commits the binding, recording patient/contact and credential versions, purpose, method, possession result, independent verifier/evidence reference, caregiver authority/consent and time. Minimize evidence storage. Issue usable enrolment challenges only against this current binding. The separate disposable setup slip shows the masked contact number; read it back in the chosen language for patient confirmation before activation and during account confirmation before funding. Readback detects mistakes/substitution and is not authentication. Funding a correctly identified beneficiary does not itself verify a contact or activate a credential.

Flag contacts matching staff numbers or reused across many unrelated patients for independent review before credential promotion. Apply bounded, privacy-safe correlation: do not expose other patients' identities, merge accounts or ban legitimate family/shared-phone use. A documented caregiver relationship can explain sharing; a staff member receiving care still needs independent approval. Contact replacement or evidenced recycling requires authorized re-verification and version invalidation of old binding/challenges/grants. Never inherit verification from a cashier edit or automatically suspend a valid binding merely because an unauthenticated person proposes a change. Recovery and protected-lane admission require the current approved binding plus their other controls.

NIST distinguishes identity proofing from subsequent authenticator binding; OWASP treats factor replacement as a sensitive recovery action. These inform the proposed separation above, not a claim of a NIST assurance level, compliant identity service or a selected provider capability. NIST identity proofing and enrolment, OWASP factor replacement.

3.5 Restore and recovery admission (AR-29/AR-30)

Restoring an old database cannot reactivate revoked caregivers/contacts, older PINs or consumed setup/reset challenges. A recovery generation outside the restored rollback domain invalidates old grants/challenges; new login or PIN setup also waits for current independent identity/authority evidence, rather than trusting restored contacts. Pause sensitive sends/bridges and new irreversible top-up/charge effects until admission/reconciliation; preserve approved urgent assistance without unverified spending authority. Recover partner cash/top-up obligations, held funds and consultation operation IDs using independently recoverable evidence or owner-approved exceptions. A missing post-backup row is not permission to credit/charge again. System15.1-15.2 and D9 gate restore acceptance before any phone pilot; they add no new M0 dependency on F14.

4. Delivery, owners and decisions

Gate / proposed owner	Deliverable and acceptance
M1 - backend, partner frontend, product/privacy, finance	Beneficiary/payer/contact identity, independent contact-binding evidence/verifier policy (section3.4), shared-phone and recovery rules, partner role/funding contracts; re-estimate M1 including financial cutover and this design work
M2 - backend, doctor frontend, clinical, finance	Reuse accepted F07/F11/F13 capacity/wallet/refund foundations; approve fixed-fee and clinical charge evidence, queue/reconnect/refund policy; no telephony expansion of M0
F14 phone pilot - backend integration owner, partner/doctor frontend owners, QA and operations	Choose P1 model before sizing; separate estimate and sandbox trial after M1/M2 acceptance. A limited pilot may join M4 only if explicitly staffed and re-estimated; otherwise schedule separately and rebase affected delivery dates
P1 funds arrangement - finance/legal/product	Select Partner-paid consultation, Consultation-only balance or General patient wallet before F14 estimate. Approve funds holder, partner collection, admitted uses, limits/reversals/fees, settlement and any authorized payment/PPI partner before live funding
P2 clinical/access - clinical/privacy/operations	Approve audio suitability and current mode/type prescribing policy, same-practitioner/prior in-person/time-window evidence and backend/UI enforcement, identity/consent, caregiver access, urgent referral, advice/referral-only charge/refund wording, reconnect and authorized outcome delivery before live calls
P3 telecom/security/operations - integration owner, security/DevOps and operations	Provider/number, callback/tariff disclosure, languages/hours, PIN transport/retention and hashing-key lifecycle (outage, rotation, compromise/re-enrolment/reset drills), private setup/current contact binding, bounded reset/recovery and staged abuse response with protected verification capacity, consented service communication, costs, bounded queues, concurrency and escalation. Security/DevOps approves key procedures; integration/operations approves provider execution and recovery evidence before release/publication

ABHA/ABDM remains a separate optional integration. Its absence does not block a patient from this channel. The existing M0 fixes remain required; this feature does not turn current wallet/recharge APIs into an approved partner collection service.

5. Required verification

* A staff-entered contact without patient confirmation cannot receive a PIN-setup callback, usable enrolment code, recovery grant or protected-lane privilege. An unprivileged, throttled possession probe does not activate any of these.
* A dishonest cashier enters their own number, copies the disposable code and answers a verification call: possession alone still cannot promote the contact or claim the patient. Require independent beneficiary evidence and a separate authorized approval actor; reject self-approval through another role.
* The disposable slip displays the masked current contact and the call reads it back; incorrect/replaced contact cannot use an old challenge. No full contact or secret enters archived financial receipts.
* Staff-contact matches and repeated unrelated-patient reuse are flagged before promotion; approved family/caregiver sharing remains usable without merging patients or revealing other identities. Test a legitimate staff-patient's independently approved exception.
* Recycled/changed contacts require fresh binding and invalidate affected old grants on authorized completion. A forged change/flag request cannot disable a legitimate patient's credential. Concurrent approval/change and callback replay enforce the same contact/credential version.
* P3 key-service outage, rotation and compromise/reset exercises have named security/DevOps and integration/operations approval; verification fails closed on unavailable keys while urgent care routing continues.
* Basic phone without app/data/SMS; shared-number patient selection, caregiver authority and PIN recovery; recycled number and guessing attacks; no history access from recharge rights.
* Synthetic PIN canary absent from provider/local logs, URL/query/path, inbox/dead letters, traces, recordings and backups after success/failure/replay. Database copy without the separate key cannot verify guesses; key outage fails closed for verification, with urgent-help routing available.

* Private enrolment and recovery: no partner account can set/read/reset a PIN; a copied printed challenge cannot claim the account without independent contact proof; expired, replayed and concurrently consumed codes fail safely. No PIN/challenge enters archived receipts, provider payload history or logs.
* Repeated setup/reset/issuance/contact-change attempts are throttled; an unauthenticated recovery request cannot invalidate a valid PIN/session. Completed change notifies registered safe contacts, revokes old grants/challenges and queues failed notifications without exposing secrets.
* One guess against each of 1,000 random patient references within the configured abuse window triggers the shared staged response across API replicas. During the controlled attack a legitimate verified-contact caller with a correct PIN still completes verification through reserved capacity; existing authorized encounters and urgent/assisted routes continue. Repeated webhook cannot reset limits or mint grants; fallback is also throttled.
* Prefunded allocation covers one top-up while two cashiers compete; replay/crash at each money boundary; wrong-beneficiary confirmation; chargeback/top-up-reversal and partner close reconciliation.
* Partner-paid consultation rejects caller-priced underpayment and unknown/invalid eligibility; authenticated duplicate returns the original result with one debit. Demonstrate each selected model's permitted uses without enabling unsupported lab/hospital wallets or substituting sponsorship for requested top-ups.
* App and phone compete for the last available funds and doctor slot; no doctor, no answer, callback failure, doctor rejection and consent withdrawal release safely without a consultation fee.
* With 500 posted and 300 held, a 300 ordinary debit fails; hold capture/release and a competing debit reconcile exactly once with no negative available balance. Values are synthetic minor-unit examples, not consultation prices.
* Duplicate/reordered answer/end events, outbound-create timeout, missing end event and worker crash: one active bridge/charge, no premature debit, bounded recovery and no repeated fee on reconnect.
* Clinical charge evidence versus telecom answer; disputed/unknown completion is reconciled rather than inferred from duration. Approved refunds compete across wallet/card/cash under one source budget.
* Audio-first versus eligible follow-up/video fixtures, same versus changed practitioner, prior in-person versus audio/video-only encounter, six-month boundary, new condition/symptoms or missing context, prohibited/unknown medicine, forged mode/type and referral-only outcome: backend denies invalid prescribing, UI explains the result and fee/refund behavior matches the announced policy.
* Set a separate concurrent-call target, call setup/queue/reconnect SLO and budget after the provider trial. The canonical 100 HTTP RPS profile remains unchanged; it does not mean 100 simultaneous calls. Measure call-control callbacks, sustained call concurrency, audio/connect quality, provider failure recovery and per-consultation cost separately.

These are unexecuted release criteria, not test results or evidence of a working helpline.

6. Primary-source constraints

Checked 8 November 2026. RBI's PPI directions distinguish own-service closed-system instruments from third-party payment use; the applicable Jansetu arrangement needs review, and merely naming a balance consultation credit does not establish an exemption. The published directions, not a draft consultation paper, are the reference. RBI PPI directions.

The telemedicine guidelines include phone/audio, practitioner/patient identity, consent, clinical suitability and records. Audio is not sufficient for every presentation; urgent cases require appropriate referral/in-person care. Cashier assistance does not make a cashier a clinician or automatically authorize clinical access. Official telemedicine guidelines.

4. Requirements and delivery roadmap

Source: documentation/planning/ROADMAP.md

Status: proposal for review, 8 November 2026. **Code baseline:** `pharmyx-fixes` at `9ca8b54606581f2158e760cad79aa7b845529184`. Existing code observations are static, not proof of production behavior. See repository evidence, performance acceptance and open source research.

Second source cross-check: the capability matrix separates existing APIs from partial and missing workflows. Low-stock queries, connected counter orders/manual payment marking, basic prescription medicine fields, in-person assisted booking and specialist payout records already exist. Preserve these; the delivery scope below includes hardening and interface completion as well as new work. Refill preferences exist, but no due-refill dispatcher was identified.

Expanded product direction: every pharmacy, lab, doctor/clinic and hospital can use Jansetu for its own in-house sales/services, billing, purchases and inventory, including business received outside the app. Marketplace participation is optional; private records and direct business remain available through the management workspace. The business operations specification defines this scope, stock/finance boundaries and acceptance. M0 duration is now re-estimate at kickoff; later planning windows are anchored to its accepted completion (T0), superseding earlier project-start calendar ranges.

Open source review: 35 projects were screened against this scope. Trial small packages/panel frameworks first (CASL, Refine Core or react-admin; conditional BullMQ; external k6), and assess ERPNext/Marley, InvenTree or OpenELIS through separately scoped adapters only where justified. Current-source license/edition exclusions and a full trial sequence are in the reuse comparison. No code was imported and no timeline saving or 100 RPS result is assumed from reuse.

Architecture review v1.7: AR-01-AR-36 disposition `(documentation/planning/ARCHITECTURE_REVIEW.md)` corrects money/wallet evidence, tenant repositories, preliminary/immutable reports, real v1 routes and recovery gates. Existing-flow payment recovery and report preservation move into M0 (F11/F12); create-only correctness indexes/readiness are F13. AR-14-AR-18 add atomic customer wallets, resume-before-retry, independent M0 recovery gates, opening balances before M2 and case-free reviewer inputs. AR-19 adds a single guarded source-payment refund budget across wallet/gateway/cash destinations; M1 is explicitly provisional and the M0 mobile confirmation change has a named owner. AR-20-AR-22 add transient/keyed PIN handling, explicit funding choices/qualified pharmacy sponsorship reuse, audio prescribing and available-versus-held guards. F14 still follows M1/M2 with no new M0 dependency. AR-23/AR-24 restore the full F14 journey and acceptance scope, private first-use PIN/bounded recovery, staged abuse continuity and explicit follow-up evidence. AR-25 adds independent patient/contact binding and masked-number/reuse checks; possession alone cannot enable credentials or protected admission. Decision rows retain PIN transport/key and prescribing mode/type with named key-lifecycle approval. AR-26-AR-36 add command/catalogue authority, quality/resource identity, restore admission/local-effect recovery, embedded-history bounds, lot price/quote rules, earlier finance contract decisions, provider/benefit recovery and precise QA gates. See the two-review disposition `(documentation/planning/PRINCIPAL_REVIEW_DISPOSITION.md)` and M0 approval packet `(documentation/planning/M0_IMPLEMENTATION_APPROVAL.md)`. These are planning changes, not implemented fixes.

Architecture package: SYSTEM_DESIGN.md `(documentation/planning/SYSTEM_DESIGN.md)` specifies the proposed module/data boundaries, tenant/patient access, transaction and payment races, API/panel lab flows, event delivery, deployment and cutover. Use the review prompt `(documentation/planning/ARCHITECTURE_REVIEW_PROMPT.md)` to challenge it against source and this roadmap before implementation. Its performance profile references the existing performance plan; v1.7 retains post-M0 windows as provisional estimates and adds separately estimated F14 phone access; v1.2 replaced fixed project-start windows with the post-M0 schedule below.

1. Product contract and evidence

Build a coordinated healthcare platform around one patient identity and a continuous care record, with daily business-management workspaces for pharmacies, labs, clinics/doctors, hospitals and service vendors. Support direct/in-house and marketplace business through the same trusted organization, clinical, billing and stock boundaries. Reuse this backend's functioning foundations and introduce missing workflows incrementally.

Sources, in order of purpose:

1. **User request:** six service areas, online/counter retail and wholesale, payments, 100 RPS, repository review, open source reuse assessment and a written roadmap before implementation.
2. **User clarification:** connected counter sales first; disconnected operation/sync later. Hospital OPD and billing first; IPD/advanced workflows later. Third-party labs with APIs use integrations; third-party labs without APIs receive a Jansetu lab partner panel.

Later scope direction: all business types need their own in-house sale/service and inventory management. Support private operations with optional public listing, both Jansetu-operated and partner businesses, and direct customer channels. Subscription/commission commercial rates remain proposed, not agreed.

3. **Product input:** `Jansetu_Master_SOP_V1.docx`, document `JS-SOP-001`, version 1.0, issued 4 November 2026, explicitly a proposed SOP. Its workflow requirements inform this plan; its illustrative priorities/timeline do not prove current capabilities or override the user's scope.
4. **Implementation evidence:** current source, route wiring, schemas and tests. Existing README/issue files are useful history and are reconciled against current code.
5. **External evidence:** primary-source project/license and standards references linked below and in the research file.

SOP input path: `/Users/shubham/Downloads/Jansetu_Master_SOP_V1.docx`. SHA-256: `dc876da03d481017d94430356bde728ba6225f0bea683ffee89b97b21bd6d286`. Content was extracted for requirements; the source document was not modified or committed. Document instructions were treated as source material, not authority to run commands, contact people or deploy software.

Priority: P0 = required before the relevant workflow handles live patients or money; P1 = staged expansion after that workflow's safe initial release; P2 = later optimization. A module cannot defer its own safety controls to a later phase. Payments move earlier than SOP section 19 because the user explicitly included them.

2. Six workstreams and acceptance

All six workstreams use the shared business foundation: individual staff, business/branch scope, private service/item masters, direct versus marketplace attribution, invoice/receipt/credit records, purchases and stock where applicable. Internal customers do not need an app login or public profile. A non-medicine supplies inventory module extends pharmacy foundations to lab/clinic/hospital supplies without removing medicine-specific safety rules. See BUSINESS_OPERATIONS.md.

U1. Pharmacy: app ordering and in-house sales

Reuse: `pharmacies`, `pharmacyMedicines`, `pharmacyInventory`, `pharmacyInventoryImports`, `pharmacyCart`, `pharmacyOrders`, `pharmacyCustomers`, `pharmacySubscriptions`, `pharmacyWallet` and existing payments. The code already represents batches, FEFO allocation, reservations, expiry/quarantine/recall states, paise totals, HSN/GST snapshots, pickup/delivery and assisted customer creation.

Deliver: pharmacy/staff verification and branch permissions; searchable catalogue; prescription upload and authorized pharmacist review; reserve -> dispense/consume or release; app cart and connected POS; printed receipt/invoice; delivery/pickup tracking; shift cash reconciliation; stock adjustments with reasons; returns into quarantine or approved restock; expiry/recall worklists; automated low-stock communication and purchasing handoff, reusing the existing low-stock query. Prescription attachment alone must not authorize dispensing: implement review, rejection reason, reviewer identity and a fulfilment gate. Do not adopt the existing guide's proposed AI approval as a substitute for authorized review.

Acceptance: two concurrent buyers competing for the last batch quantity cannot oversell; expired/recalled/quarantined stock is blocked; a rejected prescription blocks dispensing; cancellation releases the exact reservation; fulfilment consumes it once; a partial return creates linked stock and financial reversals without rewriting the original sale. The POS operator can close a shift and explain the difference between expected and counted cash.

U2. Retail and wholesale through app and counter

Keep **sales channel** (`app`, `counter`) separate from **buyer class** (`retail`, `wholesale`). Current `orderType: online|in_store` describes the channel; `buyerPharmacyId` selects wholesale pricing. Add origin channels for phone, approved import/API and direct operations through compatibility mapping, with referring party and commission attribution recorded separately. An online payment does not automatically make a direct sale a marketplace order. Add explicit entitlement/pricing policy without breaking existing fields.

Deliver: retail/wholesale price lists and effective dates; verified buyer eligibility; pack/strip/unit conversion and stock unit; quantity slabs/minimum quantities if commercially required; controlled discounts/price overrides; tax/rounding snapshots; buyer and supplier accounts; purchase order -> goods received note -> supplier invoice; purchase returns; sale returns/credit notes; receivables/payables, credit limits and aging where wholesale credit is enabled. Until credit is implemented, require settled payment rather than silently treating an unpaid order as credit.

Use the same stock authority for app and POS. Model fulfilment, invoice, collection and settlement separately. Define whether prices include tax, which entity issues the invoice, invoice numbering by legal entity/financial period, and who funds discounts. Finance review must confirm applicable GST/e-invoicing/e-waybill requirements; this plan does not prescribe rates or thresholds.

Acceptance: all four channel/buyer combinations select the server-authorized price; a retail caller cannot obtain wholesale entitlement by changing a request field; unit conversion preserves stock; approved override records actor/reason; invoice totals reconcile to receipts, refunds and outstanding balances. Connected POS can recover from a request timeout without duplicating a sale.

U3. Hospital management: OPD first

Reuse: patients/account mappings, specialists, slots/bookings, file/report access, payments and clinical records. Existing `partners` CRUD and a specialist's `hospitalOrClinic` string are not a hospital operations system.

First release: hospital organization/branches/departments, staff memberships, patient lookup and facility MRN mapping, OPD registration, appointment/walk-in token queue, triage/vitals, clinician encounter, investigations, prescriptions, service tariff, invoice, cash/online receipt, cancellation/refund, referral acceptance and discharge-document capture. An external hospital's discharge summary may be uploaded before native inpatient operations exist.

In-house business scope: private OPD services and receipts work without a marketplace listing. Add central/department supply stores, purchases/receipts, issue/receive transfers, visit/procedure consumption, stock counts, authorized waste/adjustments and cashier close. A medicine-dispensing department retains pharmacy rules. Basic hospital supplies are initial scope; they do not wait for IPD.

Later releases: admission/discharge/transfer and bed census; nursing observations and medication administration; ward billing/deposits; OT scheduling/consumables; insurance/TPA/claims if required; blood bank, radiology/PACS and biomedical inventory as separately scoped integrations. Do not put the whole list into the OPD pilot estimate.

Acceptance: an unlisted hospital can complete a private OPD visit and receipt; app and walk-in visits link to the right patient/business. Department receipts, issues/consumption/transfers and counts reconcile; users see only authorized records; charges, receipt and documents reconcile; cancellations use auditable reversals. Later IPD acceptance must prove bed transfer/census consistency and discharge reconciliation.

U4. Lab partner services: API integrations and panel

Confirmed operating model: diagnostic fulfilment includes third-party lab vendors. Support two routes according to each lab/branch's capabilities. A lab without its own APIs receives a Jansetu panel; it does not need to build an API before joining.

Daily lab business: the same workspace manages the lab's own walk-in, phone and referred customers, private test tariffs, invoices/receipts, purchases and reagent/kit/vial/consumable stock. An app account or marketplace listing is not required. Basic stock receipt, lot/expiry where applicable, consumption, waste, count/adjustment and low-stock worklists are initial scope. Origin and API/panel fulfilment mode are separate fields; prevent duplicate orders and dual stock writers when a lab also uses an external system.

Lab partner model	How orders and reports are managed	Existing foundation / delivery
API-connected third-party labs	Send orders through the provider adapter; receive status/reports through authenticated callbacks, or reconcile by polling when the provider contract requires it	Reuse LabStack/Healthians and existing external booking/catalogue/payment/report services. Add other provider adapters only against a verified contract.
Panel-managed third-party labs	Lab staff log in to Jansetu, manage assigned orders/sample status, upload reports and release them using permitted roles	Build the missing lab partner panel and its organization/branch/assignment permissions. The panel calls Jansetu's backend; no API operated by the lab is required.

One common lab-order workflow: preserve one patient/encounter link and canonical order reference, mapping existing external booking/provider IDs when applicable. Record the assigned lab/branch and `fulfillmentMode: api` or `panel` on the order. Both routes use the same permitted statuses, commercial policy, report versions, audit and doctor-review workflow. Changing fulfilment mode after dispatch requires an explicit, audited handover; never create a second provider order or payment silently when an API times out. API events and panel edits must not compete as independent writers for the same order.

API route: catalogue/serviceability/slots -> order submission -> provider reference -> status callbacks or reconciliation -> secure report ingestion -> release/review. An optional status view for API partners does not grant permission to overwrite provider-owned status. Provider outages surface to operations with retry/reconciliation state.

Panel first release:

1. Lab organization/branch onboarding, verified credentials/signatory and individual staff accounts. Lab admin, order/sample staff and report signatory have distinct permissions; a lab sees only its assigned orders and necessary patient details.
2. Configure private test menus/packages, tariffs, collection options, hours and turnaround commitments through authorized lab staff. Marketplace publication/pricing follows the agreed operations approval; neither private nor public changes alter a booked price.
3. Create direct/walk-in/referral test orders and manage Jansetu assignments in source-aware worklists: new, accepted, sample-pending, processing, report-pending and completed. Staff accept/reject with reasons and coordinate appointment/collection details; record invoice, receipt, balance and refund separately.
4. Record lab-local reference, sample collection/receipt times, rejection/recollection and processing status. Use the lab's existing barcode/reference where available; a new barcode/analyzer system is not a prerequisite for this panel.
5. Upload the original PDF/image against an assigned order; validate patient/order linkage, scan/quarantine the upload, record report date/signatory and preserve source. Authorized verification/release is separate from upload. Structured result entry is optional unless the agreed lab workflow requires it.
6. After release, update the shared timeline and create the clinician-review task and authorized patient/doctor notifications. A corrected report is a new linked version with a reason; original files remain preserved.
7. Manage supply purchasing/receipt, explicit or approved per-test consumption, waste and balances using the supplies inventory boundary. Consumption is linked to its order/procedure and cannot be posted twice by report release, callback replay or repeated clicks.
8. Show delivery failures, overdue reports and critical-result acknowledgments. Keep direct-business receipts/dues distinguishable from marketplace commissions and settlements, and extend detailed statements as the commercial scope completes.

Events and webhooks: both provider report ingestion and panel report release publish the same durable `lab.report.released` or amended-report event. These events drive patient/doctor notifications, care tasks and optional signed webhooks to subscribed systems. A panel-managed lab does not need an inbound webhook endpoint or external API to complete an order. Incoming provider events still require authentication, durable inbox/deduplication and ordering; outgoing deliveries need retry, attempt history, dead-letter handling and replay. Persist before acknowledging receipt. Validate report URL destinations/redirects and minimize clinical data in event payloads.

Safety: clinician-approved critical-result escalation with named owner, direct communication, acknowledgment deadline and backup contact applies to both routes. Delivery of a push notification is not completed clinical escalation. Clinical reviewers set thresholds and communication rules.

Optional full LIS: lack of a partner API does not require deploying a complete LIS. Basic reagent/consumable stock is now included in daily business scope. Analyzer connectivity, QC, calibration, instrument-generated consumption and accreditation workflows remain a separate lab-led decision in M8. Evaluate OpenELIS only if that deeper scope is required; verify its actual API/event extension points instead of assuming native webhooks exist.

Acceptance: complete an API-connected lab journey and a panel-managed journey, plus a direct cash-paying patient at an unlisted lab without app login. Test lab isolation, signatory release, recollection, amendments/review, critical acknowledgment and stock usage/waste reconciliation. Callback/click replay cannot duplicate booking, payment, report, consumption or refund. Failed outgoing delivery remains replayable; the panel route does not call a lab-owned API.

U5. Doctor workspace, app/walk-in bookings and payments

Reuse: specialist self-service APIs, availability/slots, booking lifecycle, prescriptions, one-level follow-up benefit, Google Meet, reminders and specialist payout calculations. This checkout supplies APIs and curl documentation, not a finished dashboard.

Already implemented medicine fields are name, dosage, frequency, duration and instructions, plus diagnosis/advice/tests and prescription PDF generation. In-person consultation mode, pharmacy-assisted booking, and admin payout/disbursal/invoice/receipt APIs also exist. Additional work covers explicit strength/route, finalization/amendments, receptionist queue, doctor finance visibility and safe integration; these existing foundations are not greenfield tasks.

Deliver: verified-credential activation; staff/delegation permissions; daily/monthly schedule, leave/blocked time and walk-in queue; booking accept/reassign/reschedule/cancel/no-show; shared capacity across app and counter; patient history subject to care assignment and consent; dated vitals/history/assessment/diagnosis; prescription strength, dose, route, frequency, duration and advice; signed/finalized encounter and amendments; test orders, report-review inbox and follow-up tasks. Preserve the existing free-follow-up policy until product decides a change; care follow-up reminders are independent of a free consultation benefit.

Clinic operations: support private appointments/walk-ins, procedure tariffs, invoices/receipts, cashier close and clinic supply purchases/usage even when public booking is disabled. Link consumables to a visit/procedure, record wastage and reversals, and test source attribution. Supply management does not by itself enable medicine dispensing; that uses the pharmacy workflow when applicable.

Money: fee/tariff snapshot, cash/UPI/gateway collection evidence, receipt, refund state, doctor earnings vs collected amount, adjustments and settlement statement. Add authenticated gateway-event recovery and scheduled gateway-to-ledger reconciliation so a closed app or missing client verification callback cannot strand an actual payment. A payout calculation or generated invoice does not prove a bank transfer occurred.

Acceptance: app and walk-in booking races cannot exceed capacity; payment timeout cannot leave an unexplained wallet debit or occupied slot; only the assigned clinician can finalize/amend a note; a second prescription version preserves the first; report review produces a recorded outcome; collections and clinician payable reconcile separately.

Basic-phone access across U1-U6

Add partner-assisted top-ups and phone consultations: an authorized pharmacy/lab/hospital identifies the patient, takes a funded account top-up and issues a receipt; the patient calls Jansetu from a basic phone, verifies identity, accepts a disclosed fee and connects to an available doctor. Reuse patient, consent, capacity, billing and clinical operations. Shared phone numbers need distinct patient selection and spending authority; a smartphone, SMS capability and ABHA are not prerequisites. Top-up rights do not grant chart access.

Use approved prefunded/confirmed funding, one consultation reservation and an evidenced fixed-fee charge; waiting/no-answer does not consume the consultation fee and a dropped call reconnects without automatic recharging. Voice-provider events are authenticated/reconciled but do not establish clinical completion. Refunds share the original source budget across destinations. F14 requires P1 selection of Partner-paid consultation, Consultation-only balance or General patient wallet before sizing, plus finance/legal/clinical/telecom decisions and a separate estimate after M1/M2; a limited M4 pilot is conditional on resources and acceptance. Do not absorb this feature into M0 or promise the existing delivery windows unchanged.

U6. Third-party vendor service workspace

Reuse: partner profile/catalogue concepts and provider adapters; pharmacy/specialist scoped APIs are useful patterns. `partners` currently has no general vendor staff/assignment/settlement model.

Labs are one vendor type. U4 is the lab-specific workspace within the shared partner/organization access model, not a second identity or isolated vendor system. The minimum staff/assignment/report panel for labs ships in M4; the broader vendor catalogue, statements and dispute experience expands in M5.

Every business can enable its internal modules before choosing marketplace participation. Include private catalogue/tariffs, direct order entry, own business invoices and stock, publication controls and SaaS plan records. Software subscription billing is distinct from recurring-medicine subscriptions. Keep private operations and marketplace finance distinguishable; commercial rates remain to be decided.

Deliver: vendor organization, branches, verified service types and documents; staff invitations, roles and revocation; service/product catalogue and price proposals; capacity/serviceability and holidays; assigned orders; accept/reject/fulfil and evidence; report upload where authorized; disputes/support; collection/commission/refund/settlement statements; payout references; webhook subscriptions and delivery logs. Third-party API clients need scoped, revocable credentials and per-client limits. Vendor types must have different permissions and forms.

Acceptance: vendor A cannot list, guess, export or mutate vendor B's orders/files; staff departure revokes access; catalogue changes cannot alter an already sold price; a vendor can complete its assigned service and reconcile its statement; no non-clinical vendor obtains the patient's full health record. Validate the first vendor types before creating a generic workflow engine.

3. SOP traceability and missing requirements

"Partial" below means related code exists; acceptance still needs validation. "Missing" means no dedicated implementation was identified in this checkout's module/schema inventory and inspected paths. Each row maps to the phase that must complete it, including controls that recur throughout delivery.

Requirement / SOP section	Baseline and required work	Delivery / priority
S01 scope and S24 patient/rural principles	Backend has accounts/bookings; connect the whole care journey and support assisted/local-language access	All phases / P0
S02 patient registration	User/dependent/pharmacy-customer models exist; add permanent patient ID, age precision/source, village/Panchayat/block/district, no-phone route, caregiver and reviewed duplicate merge	M1 / P0

Requirement / SOP section	Baseline and required work	Delivery / priority
S03 consent	No dedicated consent ledger found; versioned purpose/language/method/actor/time, withdrawal and caregiver authorization must affect access and future sends	M1 / P0
S04 EHR	Booking prescriptions/reports and specialist-local history exist; add encounters, observations, allergies/source, conditions, medications, documents and amendment history	M1-M2 / P0
S05 journey	Booking lifecycle exists; add episode-linked tasks, owner, deadlines and incomplete-care worklists across modules	M2-M4 / P0
S06 doctor onboarding/consultation	Credentials/verification fields and portal exist; enforce activation and care assignment; add complete encounter and physical-review outcomes	M0-M2 / P0
S07 prescription/medication	Name/dosage/frequency/duration/instructions, diagnosis/advice/tests and PDF generation exist; add explicit strength/route, authentication/finalization/version history and dispensing linkage	M2-M3 / P0
S08 diagnostics	External lab APIs exist; add direct/app orders, private lab billing/supply stock and no-API partner panel with shared report/signatory/review. Both routes require critical escalation	M3-M4 / P0
S09 follow-up	Booking reminders, one-level benefit and pharmacy refill preference/date fields exist; no due-refill dispatcher identified. Add refill dispatch and clinician-defined care tasks, missed-contact escalation and outcomes	M3-M4 / P0 for enabled reminders
S10 hospital referrals/emergencies	Existing `referral` is an acquisition-reward module; build clinical referral/acceptance/discharge and emergency incident/escalation separately	M4-M5 / P0 before service launch
S11 chemist operating routine	Pharmacy roles/linked customers exist; build permitted assisted registration, consent, request coordination and daily handover/reconciliation screens	M1-M3 / P0
S12 roles/access	User/admin/HR/specialist/pharmacy roles exist; add organization/branch/assignment/purpose checks, individual staff accounts, operations/field/lab/hospital/support scopes	M0-M5 / P0
S13 communication	SMS/email/push and reminders exist; add preferences, Odia/local-language templates, authorized voice-assisted workflow, delivery/retry and minimum clinical data	M1-M4 / P0
S14 money	Gateway/wallet/payout logic exists; add cash control, invoice/receipt, partial refunds, reconciliation and provider settlement statements	M2-M5 / P0 for money flows
S15 security/backups/audit	Logger and some transaction ledgers exist; remove public PHI and keys, fix auth, add purpose/access audit, retention/hold, encrypted storage and tested restore	M0 onward / P0
S16 grievances	No dedicated case-management module found; complaint ID, assignment, category, escalation, response and closure evidence	M4 / P0
S17 operations/reporting	Admin/pharmacy summaries exist; add pending-care/doctor-review, clinical incidents, village/Panchayat adoption and partner quality views	M4-M5 / P0 baseline; deeper analytics P1
S18 forms/registers	Some underlying records exist; create the explicit mappings below and printable/exportable views with permission checks	M1-M5 / P0 for released workflow
S19 priorities	Use launch gates per workflow; user brings payments forward; ABDM readiness P1, AI analytics P2	M0-M6

Requirement / SOP section	Baseline and required work	Delivery / priority
S20 KPIs	Instrument definitions/events; establish first-30-day baseline, then set business targets with clinical review	M4-M6 / P0 instrumentation
S21 timeline	SOP's 30/60/90 days is illustrative and narrower than requested commerce/HMS scope	Conditional roadmap below
S22 readiness	Patient ID/merge, assisted access, consent, timeline, verification, audit/export, backup and complaint checklist become release gates	M0-M6 / P0
S23 regulatory/standards	Track India/state, data protection, telemedicine, pharmacy/lab requirements and ABDM applicability; qualified owners approve current obligations	M0 onward / P0 for applicable controls
S25 supporting SOPs	Product/operations owners write registration, EHR, teleconsult, diagnostics, follow-up, emergency, onboarding, privacy, support and reporting runbooks	By corresponding module release / P0
S26 simulation/sign-off	Execute 10-20 synthetic journeys, including no-phone, repeat, elderly, chronic, diagnostic and emergency cases; document exceptions and sign-offs	M4 pilot and M6 rollout / P0

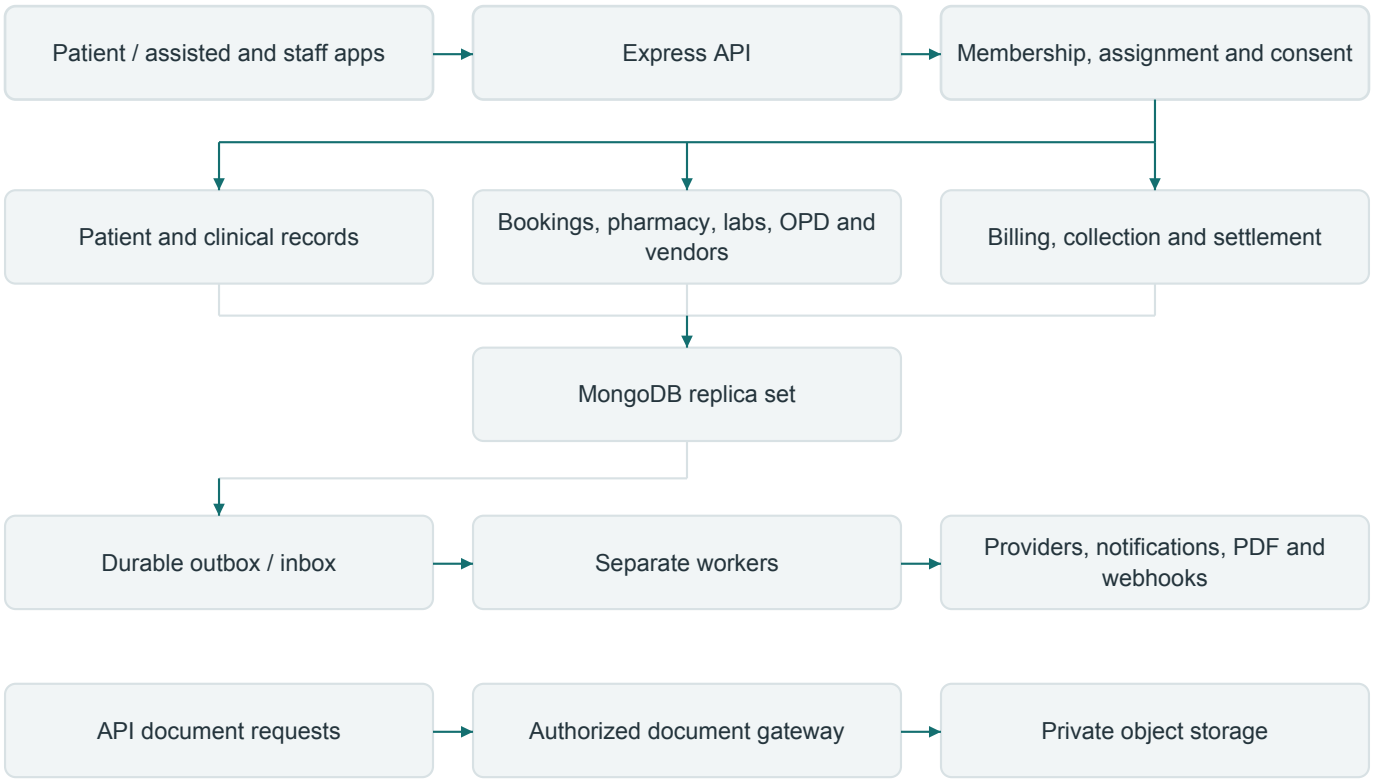
Required register mapping (S18)

SOP register	Proposed authoritative record
JS-01 registration	Patient + registration provenance + account mapping
JS-02 consent	Consent event/version and representative authority
JS-03 medical history	Clinical history, source and verification state
JS-04 consultation request	Booking/request linked to patient and episode
JS-05 consultation	Encounter with clinician-authenticated finalization/amendments
JS-06 prescription	Versioned prescription and fulfilment linkage
JS-07 diagnostic booking	Lab order and external ID mappings
JS-08 diagnostic report	Verified report versions, original document and review task
JS-09 follow-up	Care task, due date, responsible team and outcome
JS-10 hospital referral	Clinical referral, acceptance, outcome and discharge linkage
JS-11 emergency	Incident with timestamps, communications, responsible clinician and outcome
JS-12 partner onboarding	Organization/credential verification history and staff memberships
JS-13 payment	Collection/refund ledger, invoice/receipt and settlement references
JS-14 complaint	Support/grievance case with escalation and closure history
JS-15 audit	Append-only access/change/admin audit with restricted retention/export

Other necessary requirements absent from the user's short list: supplier purchasing, unit conversion, stock returns/recalls, prescription review, invoice/cash control, verified staff access, duplicate identity review, consent

withdrawal, report corrections/critical communication, wallet-versus-original-source refund policy, abandoned payment recovery, safe exports, incident handling and restore tests. Each is allocated above or in the backlog below; optional insurance/PACS/analyzer scope remains a separate decision.

4. Architecture and ownership



Retain current controllers, validation, services, repositories and models. Extract shared use cases where duplicated rules cause inconsistency; a wholesale rewrite or microservice split is unnecessary for the initial throughput target. New module names below are **proposals**, not existing directories.

Boundary / proposed module	Owns	Reuse or integration rule
`organizations` and existing `auth`	Businesses/branches/departments, staff memberships, enabled modules, publication/ownership settings and service credentials	Private operations work with marketplace disabled; `corporateId` remains a customer association, not universal tenancy
`patients`	Permanent patient identity, demographics, duplicate-review/merge history, facility/provider ID mappings	Login `user`, dependent and pharmacy customer are relationships; preserve original IDs and transaction references
`consents`	Versioned grants/withdrawals and representative scope	Access decisions use purpose, assignment, legal basis and current restrictions; a consent boolean is insufficient
`clinicalRecords`	Episodes, encounters, conditions, allergies, observations, prescriptions, report/document references	One clinical writer per artifact; original and amendments remain accessible under policy
Existing `bookings`, `slots`, `specialistPortal`	Availability, appointment holds and doctor workflow	Reuse for OPD; add staff/walk-in actions with shared capacity and payment compensation

Boundary / proposed module	Owns	Reuse or integration rule
Existing pharmacy modules + `procurement`	Medicine stock, batches, purchasing, dispensing, returns	Mongo pharmacy remains stock authority unless a separately approved ERP migration explicitly transfers ownership
`suppliesInventory` + `procurement`	Non-medicine items/units, locations/lots, supply purchases, stock movement, consumption, counts and transfers	Reuse proven primitives while preserving pharmacy safety; one writer per business/branch/item/location balance
`labOperations` + existing provider adapters	Shared lab-order/report workflow with API-connected and panel-managed partner routes	Record fulfilment mode/assigned lab per order, reuse external booking mappings, and prevent conflicting API/panel writers; full analyzer/QC remains optional
`hospitalOperations`	Facility OPD worklist, encounter context and later admissions	Reuse clinical/billing services; ERP/HMS adapter owns any external facility workflow chosen after spike
`vendorPortal`	Vendor self-service use cases, assignments and permitted statements	Calls domain services; never bypasses ownership checks or directly writes clinical/stock/financial collections
`billing` + existing `payment`, wallets, `payouts`	Invoice/credit/receipt, collection/refund records, payable/settlement references	Gateway events do not directly certify service fulfilment or payout; use integer minor units and explicit reconciliations
`businessSubscriptions`	Software plans, enabled modules/limits, billing periods, subscription states and platform billing records	Separate from medicine-refill subscriptions and business customer invoices; no presumed commission on direct orders
`careTasks`, `clinicalReferrals`, `support`	Continuity, referrals/emergency records and grievances	Keep marketing/referral rewards separate from clinical referrals
Shared document/audit/event services	File metadata, purpose/access events, delivery attempts and recovery	Existing Mongo job store may be hardened first; BullMQ is optional after a measured comparison

Access model

Authorize the actor, organization membership, branch, resource assignment and permitted purpose at the service boundary. Derive scope from trusted membership and resource data; never accept an arbitrary client-supplied tenant ID. Global patient identity does **not** grant global health-record access. Clinical sharing across organizations requires an authorized care relationship and applicable consent/legal basis. Administrative actions need explicit scope and audit; emergency access needs approved policy, reason, time limit and review.

Private patients, internal tariffs, stock and business invoices remain outside public discovery. An internal patient does not need an app account or become publicly searchable. Staff access follows clinic/lab/hospital/pharmacy roles; marketplace approval does not replace clinical verification or privacy controls.

Actor	Allowed view/actions	Boundary to test
Patient / authorized caregiver	Own or specifically authorized dependent's services and records	Family phone/account alone grants no clinical access
Chemist/counter staff	Assigned pharmacy sales, permitted demographics and prescription fulfilment	No independent diagnosis/prescription; no full EHR by default

Actor	Allowed view/actions	Boundary to test
Doctor	Assigned visits and authorized relevant history; clinical finalization	No unrelated doctor's schedule/clinical record mutation
Lab staff / signatory	Assigned lab orders/specimens and role-specific report actions	Staff upload is separate from signatory release
Hospital staff	Branch/department OPD tasks; clinical/finance views according to job	Reception or cash access does not imply clinical edit permission
Generic vendor	Own catalogue, assigned services and settlements	No unrelated patient records or other vendor's finance
Operations / field / support	Minimum necessary assistance and worklists	No unrestricted clinical downloads; privileged access audited
Platform administrator	Explicit platform/verification permissions	MFA, revocation and auditable access; no hidden clinical edits

Data and lifecycle invariants

* Separate patient identity from authentication. Patients can lack phones or use a caregiver's phone. Never merge people automatically by phone or name; duplicate candidates need authorized review, retained source IDs and reversible mappings.
* New clinical records use `patientId`, `encounterId`/`episodeId`, author, source, organization and version. Existing `userId`/`dependentId` remain during compatibility migration. Normalize timestamps to UTC; render service schedules in the branch time zone and preserve original local date/time where needed.
* Booking, payment, invoice, fulfilment, clinical review and settlement each retain their own state. Map existing enums through adapters; avoid one giant patient-status enum. Closing an episode does not deactivate the patient.
* New money calculations use integer paise and defined rounding. Snapshot price entitlement, UOM, tax policy and tariff version; historical invoices are immutable and reversals reference them.
* Stock reservation and order creation commit together. Capacity hold and booking creation commit together. Money debits and their ledger entries commit together. Provider/network operations happen outside local transactions with durable commands and compensations.
* Use unique idempotency keys scoped to actor/organization and operation plus payload hash; same key/different body returns conflict. Persist result and business reference. Keys/retention must cover the supported retry period and disconnected replay window.
* Proposed lab states: ordered -> collection scheduled -> collected -> received -> processing -> verified -> released; reject/recollect/cancel/amend are explicit branches. Doctor-review task is separate. Define allowed transitions and authorized actors before coding.
* Proposed event envelope: `eventId`, `eventType`, `schemaVersion`, `occurredAt`, resource ID/version, organization scope, correlation ID and minimal payload. Start with patient registration, consultation completion, prescription issue, report release/amendment, payment/refund and task-due events. Payload examples must use synthetic data.
* Incoming webhooks use provider-specific verification, a unique inbox key and state ordering. Outgoing webhooks use per-subscriber secret, event ID, timestamp, body signature, retry attempts and a dead-letter worklist. Secrets and full clinical payloads do not belong in ordinary logs.

Migration and rollout

1. Back up and restore into staging; inventory counts, duplicate candidates, missing patient/source references and unresolved batch information.

2. Add patient and organization mappings without removing old IDs. Treat dependents as patients; associate authorized caregivers separately. Flag ambiguous matches and retain unresolved rows for review.
3. Introduce new permissions in shadow mode with denied-access reports, then enforce by pilot organization. Never shadow-write real clinical changes to two systems.
4. Backfill timeline references using source identity and deterministic import keys. Mark imported records as historical; do not manufacture consent, clinician verification, report signatures or stock expiry data.
5. Reconcile counts, ledger totals, stock balances and document hashes. Exercise old app clients against contract tests before switching the UI.
6. Roll out by organization/branch and workflow using feature flags. Stop new writes to the affected workflow on rollback; preserve accepted orders, money and clinical records. Revert software compatibility paths rather than blindly restoring an old database over new transactions.

5. Dependency-ordered delivery

Capacity assumption: two backend engineers, two frontend engineers, one QA engineer, part-time DevOps and available domain reviewers. Existing app/admin repositories must be inspected at kickoff; none was supplied as an implementation target in this checkout. Estimates assume reuse of compatible client shells and prompt access to provider sandboxes. One backend engineer or a net-new set of client applications materially extends the timetable.

M0 is re-estimated at kickoff and ends only when its gates pass. T0 is that accepted completion, with post-M0 week 1 starting immediately afterward. Indicative downstream windows place the pilot in post-M0 weeks 11-12 and expanded rollout in weeks 19-24; earlier project-start pilot weeks 13-14/rollout weeks 21-26 and older weeks 17-20 ranges are superseded, not retained as delivery promises. Any live pilot must pass 100 RPS for its recorded released-endpoint mix; M6 repeats the gate across private business and marketplace workloads. Ranges remain conditional on the assumed team; reserve 20-30% capacity for defects, migrations and provider delays, and re-estimate after M0/M1. M1 weeks 1-3 are explicitly provisional: include legacy financial classification, finance approval and cutover/write freeze in its re-estimate, and rebase dependent windows if it extends. F14 phone scope needs its own staffed estimate before any M4 pilot commitment.

M0 includes F11 atomic wallet/payment recovery, F12 preservation and F13 verified indexes. F11 has no dependency on F07: M0 hardens existing pharmacy reservations and lab/provider payment timeouts plus cancelled/deleted money recovery; new doctor hold/capacity behavior is F07 in M2. M0 records facts/manual exceptions and preserves report versions/provider labels/current permitted recipients without waiting for expanded wallet/report policy decisions. Dates are recalculated after M0; unresolved safety defects block the affected release.

Milestone	Window / dependencies	Deliverable	Accountable roles and release gate
M0: safe/reproducible foundation	Re-estimate at kickoff; complete before T0	Resolve lock/start/build; revoke/replace tracked credentials; OTP/session/rate-limit remediation; secure and version clinical files; authenticated raw-body gateway intake, independent payment reconciliation/capture-aware expiry and recovery; ownership fixes; create-only index/readiness proof; staging/CI/audit foundation	Backend + DevOps + QA; clean install/build, auth/object-access negative tests, no public PHI or report overwrite, no bypass/embedded private keys, capture/expiry/replay and correctness-index gates pass
M1: shared business/identity	Post-M0 weeks 1-3 provisional; re-estimate including financial cutover; M0	Business/branch/staff and private/publication contracts; patient/consent, units and billing contracts; classify legacy wallet/paid/refunded evidence and reconcile opening wallet/platform-business balances before M2 posts	Backend + frontend + operations; no-phone/child and unlisted-business flows, scope denials and reviewed data contracts/imports

Milestone	Window / dependencies	Deliverable	Accountable roles and release gate
M2: doctor and business collection	Post-M0 weeks 4-6; M1 and reconciled financial cutover	Direct/app appointments and capacity, encounters/prescriptions, review tasks, business fees/invoices/receipts and approved wallet/platform-business liability map; extend M0 gateway recovery to new business/cash flows	Backend lane A + doctor frontend + clinical/finance; source/permission, signed-note, slot and money cases pass
M3: pharmacy and shared supply operations	Post-M0 weeks 4-8; M1 and M2 billing contracts; full gate after M2	Connected counter/app pharmacy, UOM/price entitlement/review; supplies inventory, purchasing/receipts, locations/transfers, clinic consumption, returns and cash close	Backend lane B + pharmacy/supplies frontend + pharmacist/accounts; stock/usage/transfer and sales/cash reconciliation; pharmacy parity tests
M4: lab business and continuity pilot	Post-M0 weeks 9-12; M2/M3	API/panel routes plus own direct lab orders, private tariffs/invoices, consumable receipt/use/waste, sample/reports/signatory, care continuity and critical escalation	Shared team + lab/clinical/operations; API/panel/unlisted direct journeys, no duplicate usage/payment, isolation, corrected report, critical acknowledgment and 100 RPS pilot mix
M5: hospital stores and business workspaces	Post-M0 weeks 13-18; M1-M4	OPD/department stores/billing; broader vendor private/public catalogue and statements; software-plan/billing records, purchasing/credit/returns and business reporting	Hospital/vendor frontend + backend + finance; private operations without listing, department transfer/count, invoice/settlement/subscription separation
M6: reliability and expanded rollout	Post-M0 weeks 19-24; released M2-M5 slices	Useful 100 RPS across direct/business/marketplace traffic, worker recovery, private data/import checks, restore/cutover, training and staged rollout	QA + DevOps + domain owners; performance, stock/money/consumption integrity, restore and documented product/clinical/operations/security sign-off
M7: disconnected sales	Separate 4-6 week estimate after core stability; M3/M6	Bounded device queue and sync, server validation, receipts/invoice policy, conflicts and reconciliation	Backend + POS frontend + QA; offline-stock/payment/replay/device-loss demonstration, then controlled pilot
M8: deeper HMS / full LIS	Scope and estimate after specialist-system spikes	Chosen IPD/nursing/OT or analyzer/QC workflows, integration ownership and operational training	Hospital/lab lead + product + engineering; estimate only after fit/gap and license/deployment decision

Parallel M2/M3 consumes the two backend lanes; later milestones do not assume the same people simultaneously deliver additional full ERP/LIS systems. An external HMS/LIS installation, customization, training and production migration can materially extend M5/M8.

M4 includes lab staff/assignment/report functions plus direct business billing and basic supply stock, using M2/M3 foundations; it does not wait for the general vendor workspace in M5. Analyzer/QC/instrument automation remains M8. Basic supplies inventory for labs/clinics/hospitals is part of the expanded initial suite, not future IPD or full LIS.

Disconnected sales design gate (M7)

Connected counter sales in M3 need timeout/retry recovery but do not need a full offline database. M7 adds only the minimum data/actions needed during an internet outage, after an explicit conflict-policy review:

* Enroll each device and operator; keep an encrypted bounded local journal with device sequence, sale UUID, actor, branch, captured price version and sync acknowledgment. A browser cache alone is insufficient recovery storage.
* Choose a stock policy before promising confirmed offline sales. Preallocate bounded stock allowances to a device/branch, or treat disconnected captures as provisional until the server accepts them. Two disconnected devices cannot independently promise the same unrestricted central balance.

* Keep cash capture distinct from gateway confirmation. Queue pending online payments; never mark them paid based on an offline screen. Define permitted offline discounts and what happens when a cached price/prescription authorization expires.
* Sync with durable idempotency and per-record versions, preserving the same sale identity across retries. Reconcile rejected/conflicting commands through an operator worklist; do not use last-write-wins for money, stock or clinical documents.
* Agree provisional receipt versus final invoice numbering and financial-period rules with finance. Never silently renumber or delete an issued document; use approved correction/reversal records.
* Test device crash/loss, journal capacity, repeated replay, partial sync, stale credentials and conflicting last-stock sales. Restrict local clinical data, enforce reauthorization on reconnect, and include device revocation/remote-risk procedures in the operational SOP.

The acceptance spike decides the supported outage duration, number of devices, local data retention, stock allowance and recovery policy. These are not required features of the first connected POS release.

Finance freeze gate (AR-33): D1/D4 merchant/funds custody, issuer/tax/rounding and payable/settlement contract decisions precede M1 common billing schema/API freeze, then M2 posting/new funds rollout. Reconciled openings remain before M2. Product/finance/legal/tax owners verify applicable obligations from current primary sources; legacy statutory section numbers in a supplied review are not adopted as current applicability. This tightening does not block M0 preservation, authorization or atomic factual money recovery.

First implementation backlog

F14 is cumulative: its original top-up, identity/caregiver, queue/capacity, one-charge/reconnect and outcome journey remains mandatory alongside security and clinical amendments. P1 funding choice cannot silently replace requested reloadable patient top-ups. The row below and PHONE_CONSULTATIONS.md sections1-5 are the joint estimate/acceptance contract; omissions or conflicting scope must be resolved before sizing.

Milestone assignment: F01-F06,F08,F11-F13,F15 are M0; F09 is M1; F07 is M2; F10 grows from M1 through M6 as its required contracts and workloads ship. F14 is a separately estimated post-M1/M2 phone integration; a conditional M4 pilot needs explicit staffing and D8/P1-P3 acceptance. Removing F07 from F11 does not waive the doctor-capacity safety gate.

ID	Work / specific existing paths	Depends on	Completion evidence
F01	Reconcile `package.json`/lockfile, pin runtime, fix `server.js`/production entry and compiled aliases; establish CI	None	Fresh isolated checkout installs, type-checks, builds and starts via documented command
F02	Rotate the three tracked service-account credentials; replace source-file loading with managed configuration; prevent future key commits	None	Old credentials disabled by owner, sanitized release artifact, secret scan; history cleanup coordinated without destructive rewrite during planning
F03	Fix `otp/services`, auth/session and production config; require MFA for sensitive staff/admin actions	F01	No public caller-selected OTP minting or unapproved authentication-email CC/BCC; legitimate server-generated login remains. No bypass, hashed bounded OTP attempts/expiry/reuse tests; missing production secrets fail startup; session revocation works
F04	Enforce ownership in `user`, dependents, booking, wallet, notification and file services; protect Meet administration endpoints	F03	Command/field allow-lists including wallet, clinical/paid state, corporate claims, doctor verification, shared medicine attributes, notifications/FCM. Table-driven own/other/role/organization tests, minimal responses and audit events

ID	Work / specific existing paths	Depends on	Completion evidence
F05	Replace public `/api/static/*` clinical access; immutable version metadata, authorization, scan and private document gateway	F04	Unauthenticated/cross-patient fetches denied, no public clinical caching; fixed-name overwrite stopped by F12; legacy-link/byte migration tested
F06	Verify incoming webhook origin; register raw-body gateway route before global JSON parser; inbox/dedupe/order checks and redaction	F01, F03; lab file processing also F05	Forged, duplicated, stale and out-of-order events tested; acknowledgement after durable persistence; valid signature does not alone establish captured money
F07	**M2 doctor capacity:** atomic appointment capacity/hold/expiry and clinician verification; composes M0 wallet/payment operation	F03, F04, F09, F11	Concurrent booking tests, timeout/retry/cancel compensation and unverified-clinician denial
F08	Separate worker lifecycle from API; job leases, retries/dead letters, outbox and process readiness	F01, F06	Crash/restart at each boundary loses no accepted work; worker lag/error metrics visible
F09	Patient/business/consent and private/public contracts; typed scoped repository/query gateway, separate patient/caregiver surface, item/unit/location and financial mappings	F04, F05, F13	Reviewed no-phone/private-business flows, membership/publication policy, stock/billing contracts and deterministic ID/import crosswalks
F10	Direct/business/marketplace datasets, stock/consumption queries, provider stubs, observability and useful 100 RPS baseline	F01, F08, F09 contracts	Reproducible scope-tagged results; no live money/provider actions; private data, stock and finance reconciliation; measured bottlenecks
F11	**M0 wallet/payment recovery:** unique wallet effect + available-funds guard (posted minus held) + $inc + movement/local state in one transaction for all reference types; no live absolute balance overwrite. Durable creating intent; reconcile/reuse active attempts before retry; webhook/poll/sweep; existing-reservation/timeout and cancelled/deleted/surplus manual recovery; common guarded captured-payment refund budget before any wallet/gateway/cash return. Owners: backend payment lead, mobile frontend owner for Confirming payment, QA/finance	F03, F06, F08, F13	Concurrent debits for funds for one; duplicate and legitimate partial refunds; crash between movement/balance writes; capture then failed verify then retry creates no second gateway order; authorized/unknown stays confirming; simultaneous full wallet credit/gateway refund on one capture permits one claim; pending external refund retains budget; bad signature, expiry/cancel/delete and replay tests; mobile verifying-error/resume contract; existing lab fulfilment intent/lookup/unknown-outcome and paid-cancel obligation tests, preserving current confirmation-pending safeguards
F12	**M0 report preservation:** replace callback stable filenames with immutable scoped hash/version manifest, keep preliminary/final/unknown provenance and exact review version; migrate available legacy bytes as history unknown	F05, F06	Partial then final/corrected callbacks preserve originals; duplicate same-version fetch does not overwrite or reopen twice; forged IDs and public cached access denied
F13	**M0 correctness-index gate:** reviewed create-only manifest job with duplicate/conflict preflight, collection/index creation completion, read-only readiness check; no automatic drop or startup reconciliation	F01	Explicitly verify wallet-effect uniqueness across reference types and pharmacy+medicine uniqueness. Missing/wrong indexes block affected writers; duplicates/conflicts fail safely; existing indexes remain; replay/numbering races pass

ID	Work / specific existing paths	Depends on	Completion evidence
F14	**Basic-phone consultation, cumulative scope:** partner-assisted funded patient top-ups and receipts; patient/caregiver verification and private PIN setup/recovery; language/IVR, bounded call queue/callback and bridge; shared doctor capacity; one consultation reservation/charge with no-answer release, reconnect and authorized outcome delivery. Retain P1 funding choice, qualified pharmacy reuse with server pricing/validation/idempotency, transient/keyed PIN and staged abuse controls, held-fund reserve/capture/release and purpose guards, and audio/prescribing/referral policy. Owners: backend integration, partner/doctor frontend, QA and operations	F07, F09, F11, F13 plus D8/P1-P3	Separate estimate after P1 selection; no new M0 dependency. Basic-phone/no-app/no-SMS journey; simultaneous funded top-ups; shared-phone/caregiver denial; queue and app/phone balance-slot races; one charge, no-answer/reconnect and authorized outcome; callback replay/timeout. Also test private single-use setup, bounded reset/notification, 1,000-reference spray with legitimate-caller continuity, PIN canary/DB-only key, server-price/underpayment/duplicate denial, purpose/held-fund races, audio-first/follow-up/prohibited prescription and referral-only cases, audio suitability and same-practitioner/in-person/time-window prescribing. Separate call-capacity/cost gates AR-25: independent patient/contact binding, masked readback and reuse flags; cashier-owned-number confirmation cannot authorize setup/recovery/protected access (phone3.4).
F15	**M0 review closure and current-path containment:** actor-specific writable fields; raw OTP-mint withdrawal and sensitive-email recipient control; wallet/booking/doctor/corporate/catalogue/notification authority; preserve quality holds in every quantity writer; current slot-resource binding; paid-cancel refund obligation or safe refusal; promotion/provider recovery rules. Reuse F03/F04/F06/F11/F12 rather than creating a new service	F01, F03, F04, F06, F11, F12, F13	Principal-review regression cases mapped to current source; forbidden actions leave state/counters unchanged and make zero provider calls; restricted lots never reactivate; release/benefit races conserve effects; no paid cancellation without a tracked obligation. Current checks are NOT_RUN; M0 acceptance needs executable evidence. New M2 capacity/M4 panel/F14 workflows are not dependencies

6. Release validation and operations

Build meaningful suites around invariants rather than one test per getter:

* **Unit/use-case:** entitlement/tax/UOM rounding, allowed transitions, consent restriction, patient mapping, medication/report amendment rules.
* **Database integration:** real replica-set stock/slot races, wallet/ledger atomicity, idempotent retries and migrations. Extend the 16 existing pharmacy cases; do not count skips as execution.
* **API authorization/contract:** every relevant role, unrelated patient, organization, branch and provider identifier; backward-compatible app contracts and versioned integration payloads.
* **End-to-end:** unlisted pharmacy/lab/clinic/hospital daily operations without patient app login; app/counter/phone origins and referral attribution; supply receipt/consumption/waste/transfer/count; correct private/public tariffs and separate subscription/commission; counter retail/wholesale, no-phone/caregiver care, refund/return, API/panel lab reports, hospital OPD and vendor settlement; local-language/print and interrupted connectivity.
* **Failure and load:** provider timeout, duplicate callbacks, stale events, crash after debit or before enqueue, worker restart, storage failure and restore; 100 RPS workload. Payment sandboxes/stubs only.

SOP S26 requires 10-20 synthetic patient journeys and named product, clinical, operations and technology/security sign-off. Include an emergency escalation drill approved by clinical operations, without sending false emergency communications. Each gate needs an artifact: run identifier, deployed commit, dataset/hardware, test result, reconciliation and unresolved exceptions with owner. Core access, patient safety, lost-money and stock corruption failures block release.

Operational dashboards must show unassigned requests/wait time, pending reports and clinician reviews, overdue care tasks, prescription review backlog, critical-result acknowledgments, failed/refunded/unsettled collections, stuck imports/jobs/webhooks and complaints. Log patient-record reads/downloads, clinical amendments, consent changes, exports, privilege changes and financial overrides with actor, purpose, resource, timestamp, outcome and correlation ID. Audit content must be minimized and access-controlled.

Measure SOP KPIs with explicit denominators: registration attempts/completions, requests/accepted/completed consultations, request-to-acceptance duration, prescribed/completed investigations, collection-to-report duration, due/completed follow-ups, initiated/completed referrals and episodes due/closed. Establish a 30-day baseline before setting clinical/business performance targets. Technical SLOs can be set before launch and validated in staging.

7. Standards and operational decisions

For interoperability, plan mappings to Patient, Encounter, Observation, Condition, AllergyIntolerance, MedicationRequest, ServiceRequest, Specimen, DiagnosticReport, DocumentReference, Composition/Bundle and invoice artifacts as applicable. Use the published ABDM profiles and sandbox requirements when the integration begins; internal JSON documents alone do not establish ABDM conformity. ABHA remains optional in registration. Official ABDM FHIR guide.

The SOP's Indian data protection reference needs a current applicability/commencement register. MeitY publishes the DPDP Rules 2025, commencement timeline and corrigendum; the official announcement describes phased implementation. Do not assume every obligation became effective on the notification date. Product/privacy owners must record the applicable provisions, dates, data roles and deployment controls for each release. MeitY official instruments, official announcement.

Clinical owners must validate teleconsultation consent, practitioner eligibility, prescribing and physical/emergency referral protocols against the current professional framework. Pharmacy/lab/hospital owners must validate their licensing and state-specific operating requirements. This is a delivery gate with named owners, not a claim that the code is compliant. MoHFW telemedicine guidance reference. The existing HIPAA remediation note does not by itself make US HIPAA the governing framework for this India-focused product.

Decisions to resolve at kickoff; planning continues under these assumptions

Decision / owner	Working assumption and effect
Brand and legal entity - product/finance	Jansetu is the product roadmap; EWA/Myewacare identifiers remain until a separate branding/API migration
Launch geography, organizations, branches and volume - operations	Controlled single-region pilot with several partner types; validate tenant count, peak-hours and data residency/vendor contracts before deployment
Exact vendor categories - product/operations	First pharmacy, lab and hospital service partners; broad non-health perks/marketplace workflows need their own scope
In-house operations and publication - product/business owners	Private daily sales/services, billing and inventory for all four business types; optional marketplace listing, business ownership and source attribution
Software subscription and marketplace fees - product/finance	Proposed separate revenue streams; no assumed commission on direct business. Decide plans/rates, eligibility, billing entity and settlement terms
Existing platform collection, capture and ledger map - finance/product/backend	Document configured merchant/capture mode in M0; no new fund route or automated payouts; fix D1/D2 account map before M2 receipts/subledger
App/admin client code and team capacity - engineering	Reuse compatible shells where available; inspect client repositories before assigning dashboard estimates
Wholesale eligibility, UOM/slabs, credit and invoice entity - pharmacist/finance	Licensed/verified buyer rules and configured prices; prepaid until credit controls pass acceptance
Refund destination, fees and settlement rules - finance/product	Record evidenced legacy wallet credits as wallet liability, not original-method gateway refunds; decide wallet scope/refund destination and account for platform-business dues before M2 posting automation
Lab partner route and depth - lab lead	API or panel plus direct lab orders, billing and basic reagent/consumable stock; signatory/review required, analyzer/QC/instrument automation only if selected
HMS operating depth - hospital lead	User confirmed OPD/billing first; inpatient workflows remain a separate estimate
Emergency/critical escalation, after-hours cover - clinical/operations	Named accountable clinician and backup communication required before the feature launches
Hosting budget, RPO/RTO and operating SLOs - product/DevOps with security/finance/clinical	Proposed starting targets; per-effect irreversible recovery and restore authority admission D9 must be approved/proved before pilot; resize using measured bottlenecks
ABDM/ABHA/insurance requirements - product/privacy/clinical	Optional ABHA and standards-ready records now; sandbox/production onboarding separately scheduled
Basic-phone funding/call policy - finance/legal/clinical/privacy, identity, security/DevOps, integration/operations	D8/P1-P3: choose Partner-paid consultation, Consultation-only balance or General patient wallet before F14 sizing; approve funds holder, authorized partner collections, admitted uses/limits, settlement/reconciliation, fees/refunds and any authorized payment partner. Also approve private PIN setup/bounded recovery, independent contact binding, PIN transport/key lifecycle (outage, rotation, compromise/reset), staged abuse continuity, audio suitability, prescribing mode/type/follow-up evidence, clinical charge, identity/consent and telecom recovery before live F14. M1 identity owners approve binding evidence; P2 clinical approves prescribing; P3 security/DevOps approves key procedures with integration/operations provider/recovery acceptance
Open source license and support model - engineering/product	No wholesale code import by default; approve pinned components after fit, license, security and upgrade-cost review

5. Repository evidence and launch blockers

Source: documentation/planning/REPOSITORY_REVIEW.md

Reviewed: 8 November 2026. **Repository:** `/Users/shubham/Desktop/SwasthaDisha/ewa_backend`. **Branch/commit:** `pharmyx-fixes` / `9ca8b54606581f2158e760cad79aa7b845529184`. The working tree was clean before planning documents were added.

Second source cross-check: CAPABILITY_MATRIX.md distinguishes existing APIs, partial workflows, missing modules and runtime evidence not available. It clarifies that low-stock queries, counter cash marking, basic prescription fields, assisted in-person booking and specialist disbursal records already exist. Refill-date preferences do not establish an implemented reminder dispatcher.

Lab scope clarification: the existing third-party API integrations remain part of the product. Labs without APIs receive a Jansetu partner panel for assigned orders/sample status/report release; that panel is missing in this checkout. A complete LIS/analyzer system is optional later scope, not the default response to a partner lacking APIs.

Expanded business scope: every pharmacy, lab, clinic/doctor and hospital must manage direct in-house services/sales, billing and inventory, with optional marketplace publication. Current stock classes/models are pharmacy/medicine-specific and refill subscriptions are not SaaS plans. New supplies/consumption, standalone-business and software-plan requirements are tracked separately in BUSINESS_OPERATIONS.md and the nine added capability checks; no underlying code capability was changed by this scope update.

Scope and method

Used the codebase-memory graph for architecture, symbol discovery, call tracing and source snippets, followed by source/schema/route reads and a Git-tracked inventory. The graph's recorded head matched the working checkout. Counts below describe the baseline before these planning documents:

Inventory	Result
Tracked files	436
TypeScript files under `src1`	354
TypeScript lines under `src1`	62,920, including comments/blank lines
Immediate feature/support directories under `src1/modules`	53
Module route files	45, plus central route composition
Pharmacy database integration cases	16; conditional on dedicated test MongoDB URI
Tracked CI, Docker/Compose/Terraform/process-manager definitions found	None in this checkout
Root license file found	None; package metadata declares ISC, which does not establish external-module license compatibility

Every module is represented in the inventory below. Depth differs: critical identity/access, order/stock, booking/payment, prescription/report and job paths were read in detail; peripheral content/support modules were reviewed structurally through their symbols, routes and schemas. This is a repository-wide planning review, not a line-by-line proof of all

63,000 lines or a penetration test. Current client applications, production settings, database contents and infrastructure were not available in this checkout.

Documentation consulted includes the TypeScript guide, pharmacy product/API/hardening guides and issue lists, automation overview, specialist flow designs/curls, upload flow, LabStack/Healthians notes, health-assessment guide, HIPAA remediation note, test scripts and API collections. Historical docs are not accepted as proof that an issue is fixed or a proposed feature exists. In particular, the HIPAA note references an older OTP literal and a webhook middleware path absent from this checkout; the current findings below use current code instead.

Archives were enumerated without extraction or execution: `Archive.zip` contains 364 entries primarily under `.git`; `remaining.tar.gz` contains 209 entries under files/uploads plus a credential file/log. No application `.ts`/`.js` source entries were identified in either archive. Uploaded documents, logs, sample data and stored API results were treated as potentially sensitive artifacts, not current test evidence. Their contents were not copied into this plan or exhaustively reviewed as patient records.

What can be retained

Area	Implementation evidence	Practical limit
Layered backend	Express/TypeDI controllers, services, repositories, Joi validation and Mongoose models in `src1`	Good extension point; inconsistent authorization must be fixed before adding more roles
Pharmacy catalogue and stock	Inventory model `(src1/modules/pharmacyInventory/model/index.ts)`, reservation repository `(src1/modules/pharmacyInventory/repository/index.ts)`, `reserveBatches`	Embedded batches, FEFO, expiry/status filtering and transaction-based reservation exist; full procurement, returns and regulatory dispensing workflow are not demonstrated
Shared daily-business foundations	Pharmacy stock/prices/imports, booking fees, manual paid marking and scoped specialist/pharmacy roles	Reuse patterns and proven safety rules; no general lab/clinic/hospital supplies ledger, consumption, department transfer, private/public business settings or SaaS plan domain was identified
App and counter orders	Order service `(src1/modules/pharmacyOrders/services/index.ts)`, `create`, `resolveOrderItemsPricing`, cart routes `(src1/modules/pharmacyCart/routes/index.ts)`	Current code has online/in-store channels and customer/pharmacy buyer distinctions; this is not proof of a complete POS application
Money calculations	Order paise fields, HSN/GST snapshots; payment verification `(src1/modules/payment/services/index.ts)`, `verifyAndMarkPaid`	Reuse signature/payment checks; collection, invoice, refunds and settlement still require unified reconciliation and failure tests
Doctor workspace API	Specialist routes `(src1/modules/specialists/routes/index.ts)`, portal service `(src1/modules/specialistPortal/services/index.ts)`	Profile/slots/appointments/status/prescription/history exist; no full dashboard client or unified EHR here
External diagnostics	External booking service `(src1/modules/externalBookings/services/index.ts)`, LabStack and Healthians clients	Supports provider booking/report coordination; native accession, specimens, lab signatory and QC are separate capabilities
Notifications and asynchronous records	Existing notification idempotency, persisted jobs and pharmacy side-effect state	Useful foundations; worker draining, delivery/retry and outbox durability need work
Assisted pharmacy users	Pharmacy customer service `(src1/modules/pharmacyCustomers/services/index.ts)`, `createCustomer`	Reuses a user by phone and creates a pharmacy relationship; still assumes a phone and cannot substitute for a master patient identity

Findings that determine delivery order

Priorities here refer to release readiness, not a claim of exploitation. Fixes are proposed; **no application fixes were made in this review**.

ID / priority	Observed evidence and consequence	Remediation / validation
R01 / P0	Three tracked root JSON files are service accounts with non-empty private-key fields: the two `ewa-health-care-firebase-adminsdk-*.json` files and `hybrid-flame-469909-r0-e32a5b65e2b9.json`. Credential values were not printed or reproduced.	Authorized owners revoke/rotate keys, replace file-based secret distribution, sanitize release artifacts and add secret scanning. Removal from the working tree alone does not revoke a credential. Coordinate any history cleanup separately.
R02 / P0	OTP service `(src1/modules/otp/services/index.ts)`, `verifyOTP`, accepts a fixed bypass and skips expiry for it. OTPs are stored directly; the OTP router is publicly mounted.	Remove bypass, secure challenge creation/verification, hash challenges, bound attempts/resends/expiry and test reuse/rate limits. Do not retain a production bypass for testing.
R03 / P0	Route composition `(src1/routes/index.ts)` exposes `/api/static/*` without authentication. File controller `(src1/modules/file/controller/index.ts)`, `serveFileByPath`, reads local files synchronously and sets one-year public caching. Clinical prescriptions/reports use this namespace.	Separate public marketing media from private clinical documents; object/consent authorization, private/no-store policy, private storage and auditable expiring access. Retire legacy public links through a migration.
R04 / P0	User routes `(src1/modules/user/routes/index.ts)` inherit authentication but no general ownership guard. User controller `(src1/modules/user/controller/index.ts)`, `ensureHRCanAccessUser`, restricts HR only and returns early for other roles; CRUD passes caller-supplied IDs into the service.	Deny unauthorized list/read/update/delete across users and staff. Audit adjacent dependents/bookings/wallet/notifications/file paths using a complete actor/resource matrix; do not assume authentication supplies authorization.
R05 / P0	Webhook routes `(src1/modules/webhooks/routes/index.ts)` have payload validation but no origin authentication/signature check. Webhook service `(src1/modules/webhooks/services/index.ts)` calls `syncBookingFromWebhook`, which can change booking state, fetch reports and trigger wallet refunds. Raw headers/body are stored.	Authenticate per provider contract; durable inbox, dedupe and ordering, redact secrets and minimize retained payloads, asynchronous processing and replay tests. Confirm edge protections separately; none are established by this checkout.
R06 / P0	Configuration `(src1/config/config.ts)` has a fallback JWT secret and validates only MongoDB URL in production. Meet routes `(src1/modules/meet/routes/index.ts)` expose OAuth administration/test endpoints without access guards.	Fail startup without required secrets, restrict administration/test routes, use state-bound OAuth callbacks, session rotation/revocation and staff MFA. Verify external effects only in a sandbox.
R07 / P0 before clinic expansion	Booking creation `(src1/modules/bookings/services/index.ts)`, `createSpecialistBooking`, saves the booking before separately incrementing slot count; increment errors are logged and ignored. Slot repository `(src1/modules/slots/repository/index.ts)`, `incrementBookingCount`, increments without an atomic capacity predicate. Follow-up benefit, wallet and booking writes also cross failure boundaries.	Atomic capacity hold + booking, scoped idempotency and explicit compensation/reconciliation for failed payments/benefit use. Test concurrent app/walk-in claims and failures after each write.
R08 / P0 before p rescribing/dispens ing	Order service `(src1/modules/pharmacyOrders/services/index.ts)`, `assertPrescriptionProvided`, establishes attachment/pending state; no pharmacist approval endpoint is exposed in the order router. Portal prescription `(src1/modules/specialistPortal/services/index.ts)`, `createAppointmentPrescription`, updates the booking prescription in place. Booking creation checks clinician activity but no verification gate in that inspected path.	Authorized review and fulfilment gate; clinician verification enforcement; signed/finalized clinical versions with amendment trail, minimum necessary pharmacy data and complete medication details.

ID / priority	Observed evidence and consequence	Remediation / validation
R09 / P0 build gate	Clean `npm ci` fails: lockfile is missing `xlsx@0.18.5` and its dependencies. `package.json (package.json)` starts legacy `server.js (server.js)`, whose imports target absent `src/`; maintained code lives in `src1/`.	Repair/review lockfile, pin runtime, establish one supported production build/start command and compiled alias resolution; verify on a clean isolated checkout.
R10 / P1 platform prerequisite	Queue `(src1/modules/queue/service/TaskQueue.ts)` schedules every ten seconds. Consumer `(src1/modules/queue/service/JobConsumer.ts)` claims one due job per `init`, with a 120-second lease and process-local running flag. `publishNow` can await consumption inline. No explicit dead-letter/retry-count/heartbeat model is present.	Dedicated workers, bounded draining/concurrency, lease recovery, retry/backoff/dead letters, event durability and queue-lag monitoring. Retain idempotent business effects. Compare a hardened Mongo worker with BullMQ only if justified.
R11 / P1 clinical foundation	User `(src1/modules/user/model/index.ts)` and dependent `(src1/modules/dependents/model/index.ts)` are account/family models. No dedicated consent, master patient, encounter, clinical referral, support case or full lab/hospital operations module was identified. Specialist history is scoped to that specialist's bookings.	Patient/account mapping, consent/authorization, encounter and task model. Add clinical referrals separately from acquisition rewards. Keep historical IDs and payment links during migration.
R12 / P1 runtime/scale	`src1/index.ts (src1/index.ts)` listens before the asynchronous DB/backfill initialization finishes. `/health` always returns process success. Schedulers/backfill start inside each API process; file/PDF work uses local disk.	Readiness and graceful SIGTERM; deploy-time migration ownership; separate workers; private shared object storage and centralized observability. Test two instances and worker restart.
R13 / P1 operational completeness	No tracked CI/deployment/load-test definition found. `npm test` intentionally fails; `test:api`/curl/Postman results are not a hermetic regression suite. Pharmacy integration tests skip without a test replica-set URI.	CI with type/build, real integration and role/contract tests; safe provider stubs; repeatable load/restore harness. Preserve artifacts distinguishing passed, failed, blocked and skipped.
R14 / P0 money recovery	Payment route is client `/api/v1/payments/verify`; `validateSignature` persists failed on bad input, `handleGatewayTerminalStatus` accepts authorized/captured, `createOrder` calls gateway before local save, cancelled/missing contexts are not a safe recovery boundary. Native 35-minute abandonment has no independent gateway reconciliation. Inspected cancellation refunds credit customer wallets and mark refunded.	M0 F11: raw-body webhook plus independent polling/sweep, captured-only collection, durable creating attempt, gateway-aware bounded expiry and cancelled/deleted/surplus-payment recovery. Do not treat wallet credit as gateway refund; inspect actual deployed capture settings separately.
R15 / P0 report integrity	`persistReportsOnDeliveredWebhook` writes stable `diagnostics-report-{id}`/Healthians filenames through `persistDiagnosticsReportPdf`, retaining the same public URL; Healthians allows partial reports. Later bytes can replace evidence a clinician already reviewed.	M0 F12/F05: immutable tenant/order/version-scoped bytes and manifest, explicit preliminary/final/unknown provenance, exact-version review and corrected-release task. Legacy overwritten history is unknown, not reconstructed from assumptions.
R16 / P1 before affected pharmacy release	Pharmacy keeps embedded batches but eligible expiry compares string dates to UTC today. A local-day expiry policy can differ at the boundary; new per-lot supply indexes cannot be applied as if existing pharmacy were already per-lot.	Preserve embedded medicine batches; define pharmacist-approved local expiry/precision and test reserve/consume/release/list boundaries. M3 parity and any current-path correction precede affected live release.
R17 / P1 customer-wallet atomicity	Seven cited booking/external/payment/wallet paths read balance, insert movement and overwrite absolute balance separately; customer refund uniqueness is scoped only to pharmacy_order in the current schema. Concurrent spending/refunds or crash gaps can disagree with movement history. Production index presence and occurrence frequency were not verified.	M0 F11/F13: one customer-wallet operation with stable effect identity across reference types, guarded debit/atomic credit plus movement/local state/outbox in one transaction. Verify wallet-effect and pharmacy+medicine indexes; test concurrent debits, duplicate and legitimate partial refunds, crash rollback and missing-index refusal.

These additions follow the supplied architecture review and disposition `(documentation/planning/ARCHITECTURE_REVIEW.md)`. Existing gateway lookup occurs inside client verification; the gap is independent recovery, not absence of every gateway fetch. Runtime balances/provider settings and actual incident frequency were not observed.

Existing pharmacy fixes deserve retention: transactional FEFO reservation/order creation, exact allocation release/consumption, archive rather than physical deletes, import-row recovery and payment/notification idempotency appear in source. These should be tested and extended, not described as absent because an older issue file listed them.

AR-21 source clarification: PharmacyWalletService.payConsultation is an existing unrouted sponsorship primitive, not the requested patient top-up/phone feature. Its transaction and conditional paid-state write protect against competing paid transitions; the review's blanket absence-of-duplicate-protection claim is too broad. It still trusts the supplied amount and has no dedicated exposed validation/idempotent-result contract. Retain the guard and keep the method unexposed pending F14/P1 and M1 identity work.

All module dispositions

Counts are tracked TypeScript files, not maturity scores. "Extend" includes verification/hardening of existing code. Modules sharing a row are still separately named and counted. Core rows received detailed flow reads; peripheral rows are structural review and need normal implementation-time validation.

Module	TS files	Current responsibility / proposed disposition
`addresses`	7	User addresses/default location; extend ownership and rural administrative geography
`admin`	7	Administrator records; individual staff/MFA/permission review
`appVersions`	7	Version policy; retain for client compatibility and staged rollout
`auth`	6	Role login/OTP/JWT; harden and add membership/session lifecycle
`banners`	7	Marketing content; retain, restrict authoring, separate public media from clinical files
`blogs`	7	Authenticated management/public content; retain with editorial permissions
`bookings`	13	Internal consultation/packages and provider clients; retain, isolate adapters and fix slot/money failure boundaries
`camps`	6	External camp discovery/order/bulk flows; retain with organization scope and safe bulk retries
`common`	12	Validation, repositories, error/query/date helpers; retain without assuming generic repository methods enforce authorization
`config`	3	Module configuration service/types; reconcile with root `src1/config` and settings
`corporates`	8	Corporate customer association; retain, do not repurpose as universal vendor tenancy
`coupon`	7	Discount eligibility/usage; retain and test concurrent redemption and money rounding
`dashboard`	5	Admin/operations summaries including pharmacy health; extend to pending-care and clinical/financial operations
`dependents`	7	Family/dependent records; map each person to patient identity and representative authorization
`email`	1	Email delivery; preserve templates, add consent-aware durable retries/minimal content
`externalBookings`	12	External order/payment/status/report orchestration; retain adapters, harden inbox, retries, report access and reconciliation

Module	TS files	Current responsibility / proposed disposition
externalCatalogItems	7	Provider catalogue overrides/metadata; add scoped vendor proposals/versioned sold-price snapshots
fcm	7	Device/token registration; review ownership, minimize push data and lifecycle/revocation
file	5	Generic upload/local download; replace clinical public-file model with controlled document access
healthAssessments	9	Questionnaires, scoring and recommendations; separate clinical governance/decision support from core release scope
healthTips	7	Health content; retain, review publication roles and clinical content ownership
hr	7	Corporate HR operations; retain limited corporate scope, prevent general clinical privilege
meet	5	Google OAuth/Meet integration/testing; restrict administration and move slow creation into managed work
middleware	3	Authentication/role guards/error handling; add consistent resource/membership checks
news	7	News content; retain and restrict authoring
notification	7	In-app/push notifications and some idempotency; extend preferences, event retries and care-task links
orderHealthChecks	7	Pharmacy-linked health observations; link source/author/patient to clinical record without chemist diagnosis
otp	7	OTP persistence/verification; remove bypass and harden challenge lifecycle
partners	7	Partner profile/type/verification CRUD; extend through organization/staff/assignment services
patientTracking	3	Pharmacy-linked orders/subscriptions/health-check summary; retain scoped view, not a longitudinal EHR
payment	7	Gateway records/verification/context processing/refunds; retain checks, extend robust reconciliation and cash/refund policy
payouts	7	Specialist billing/disbursal records and invoice/receipt assets; distinguish calculation, payable and confirmed settlement
pharmacies	7	Pharmacy onboarding/profiles/verification and public discovery; add branches/individual staff/licence lifecycle
pharmacyCart	7	Actor-scoped cart and checkout; retain fresh pricing checks and stock transaction boundary
pharmacyCustomers	7	Pharmacy-to-user relation and assisted creation; replace phone-as-identity assumption with patient mapping
pharmacyInventory	7	Embedded batch stock, allocation and adjustments; retain, extend purchasing/returns/UOM and measured query performance
pharmacyInventoryImports	6	Preview/commit/retry/error download and cleanup; retain recovery, move large work off API
pharmacyMedicines	8	Catalogue, tax/prescription flags, discovery/name migration; retain and govern master data/units

Module	TS files	Current responsibility / proposed disposition
`pharmacyOrders`	9	Online/in-store, retail/pharmacy buyers, payments and side effects; extend invoices/review/returns/shift closure
`pharmacySubscriptions`	7	Recurring medicine/refill records and reminder preferences; retain, add missing due-date dispatch and separate clinician follow-up
`pharmacyTracking`	3	Pharmacy sales/customer/subscription summaries; retain with branch/vendor scope and rollups if needed
`pharmacyWallet`	7	Prepaid pharmacy wallet/ledger/top-up and atomic helpers; preserve transaction/idempotency and reconcile to collection/settlement. Unrouted payConsultation transaction conditionally guards booking paid but trusts req.amount. Before exposure require authoritative server fee, payment validation/durable idempotent result and eligibility; no lab/hospital funding or phone-only flow is established
`queue`	7	Persisted scheduled jobs/consumers; harden worker execution and recovery before scale-out
`referral`	5	Acquisition codes/reward earnings; retain separately from new clinical-referral domain
`settings`	3	Settings types/services; consolidate authoritative config ownership and production validation
`slots`	7	Specialist/package capacity; add atomic holds, doctor/branch ownership and app/walk-in race tests
`sms`	1	SMS delivery; add consent/local-language/delivery control and abuse protection
`specialistPortal`	2	Doctor-facing use cases/validation; extend walk-ins, shared EHR, review, note versions and finance visibility
`specialists`	7	Credentials, self routes and statistics; enforce verified activation/assignment and staff delegation
`user`	7	Account/demographics/corporate association; fix object access, keep account distinct from patient identity
`wallet`	7	User wallet/transactions/recharge; verify all debit/credit/refund failure boundaries and ownership
`webhooks`	7	Raw callback storage and booking sync; authenticate, dedupe/order and decouple processing
`wellnessPackages`	7	Internal service/packages; retain governed catalogue/tariff and appointment compatibility

Supporting paths reviewed: `src1/index.ts`, central routes, config/container, DB connection, logger, Firebase integration structure, index synchronization and batch migration scripts, root package/TypeScript/environment setup and test/documentation assets. No private keys, patient payloads or log contents are included in this report.

Verification record and practical limits

Check	Observed result
Git and graph baseline	Same commit; working tree initially clean
Dependency installation, restricted environment	DNS resolution blocked registry access on initial attempt
Dependency installation with network access	`npm ci --ignore-scripts --no-audit --no-fund` failed with EUSAGE: missing `xlsx` and transitive entries in lockfile
`npm run type-check`	Blocked: `tsc: command not found` after failed install; no compilation conclusion
`npm run test:pharmacy`	Blocked: `tsx: command not found`; none of the 16 cases executed
Database/provider/payment/live API checks	Not run; no production endpoints or databases were exercised
Load/failover/restore checks	Not run; 100 RPS, availability and recovery targets remain unverified
Source changes	Documentation only; no secret rotation, migration, source import or security fix performed

The findings are enough to sequence implementation, not to approve production use. Re-run the baseline after F01 and retain actual test results. The roadmap maps every blocker and requested service area to delivery work and acceptance.

6. 100 RPS and recovery acceptance plan

Source: documentation/planning/PERFORMANCE_PLAN.md

Status: proposed acceptance plan, 8 November 2026. No benchmark was run. Current install/type/test blockers are in the repository review. These targets are engineering assumptions for review, not measured capacity or a hardware guarantee.

Expanded product scope includes daily business operations: private customers/orders, lab/clinic/hospital supplies, invoices/receipts, transfers/consumption and optional marketplace listing. Keep the target at 100 aggregate RPS; broaden the useful-work mix and integrity assertions instead of assuming app-only results cover private business operations.

Define the target precisely

Target **100 aggregate incoming HTTP requests per second** across the Jansetu API under a representative mixed workload, with correct stock, money, slot and clinical behavior. This is neither 100 concurrent users nor 100 completed patient journeys/s. At 100 RPS, 1,000 ms average response time implies approximately 100 requests in flight; size the generator and application from actual latency, not user-count guesses.

Measure achieved request rate, latency, unexpected failures, scheduled/dropped work and business outcomes. Count every measured incoming HTTP request; browser preflight, polling, retry and webhook traffic must be included if present in the selected workload. Setup/token seeding is outside the steady-state measurement and separately reported. Provider API calls and object-storage bytes are separate downstream metrics. Whole-file transfers, PDF generation and bulk import/export get separate resource tests.

Initial interpretation is **100 RPS total**, not 100 per endpoint, per organization or per service. A later requirement for 100 successful writes/s or 100 fully external provider transactions/s needs its own capacity and provider-quota test.

Lab traffic includes both API-connected third-party labs and panel-managed third-party labs without provider APIs. Count lab-panel worklist/status/report commands as incoming Jansetu API traffic. Provider callbacks apply to API partners; panel report release creates the same downstream event without a provider callback. No lab-owned API is required for the panel scenario.

Proposed acceptance gates

Measure	Proposed release threshold	Measurement
Offered/achieved throughput	Schedule 100 requests/s; at least 99 requests/s completed over steady state, with no dropped scheduled iterations	Report offered, started, completed and successful rates separately, plus per-minute buckets
Simple/list reads	p95 <= 500 ms, p99 <= 1,500 ms	Per endpoint class, bounded page/body sizes and representative filters
Local transactional commands	p95 <= 1,000 ms, p99 <= 2,000 ms	Booking/stock/receipt writes with provider responses stubbed according to the test contract
Incoming event acknowledgment	p95 <= 300 ms after authentication and durable inbox persistence	Heavy processing tracked separately; do not acknowledge before persistence
Unexpected HTTP/transport errors	< 0.1% during the 100 RPS steady run	Include 5xx, unexpected 4xx, timeouts and failed transport; report intentional conflict/denial tests separately
Business integrity	Zero oversells, overbookings, unexplained wallet/ledger differences, duplicate financial effects or unauthorized disclosures	Reconcile database and event counts after each run
Background processing	Normal-task age p95 < 30 s; no continuously growing backlog at steady load	Submission-to-start and submission-to-completion; clinical escalation deadlines set separately by clinical lead
Headroom	No sustained resource/pool saturation or monotonic memory growth; queue drains after load	CPU, RSS/heap, event-loop delay, connection waits, transaction retries, disk/network, queue age
Recovery	No loss of durably accepted commands; restart safely replays work	Process crash/restart with before/after reconciliation

Also record offered/achieved cache-hit and miss ratios per category against a fixed warm/cold acceptance profile and agreed maximum cache ratio; cache-heavy discovery cannot replace required successful local writes.

These thresholds apply to the specified workload and deployment. Do not hide one slow endpoint behind a global percentile. Purposeful 409 stock/slot conflicts and 403 access denials are correctness cases; a load run dominated by them does not prove useful throughput.

Provider-dependent booking/report latency needs a separate SLO using observed provider distributions and quotas. Where a durable command is accepted asynchronously, return an explicit processing state and expose its outcome; a fast HTTP acknowledgment is not a completed external booking.

Workload and synthetic data

Start with this **100-request/s** expanded mix, then replace weights using pilot traffic. Tag source channel and business type; at least 50 requests/s in this proposed mixed profile should exercise private/direct business operations, including successful writes. Marketplace listing must be disabled for some test organizations. An additional marketplace-focused profile also runs at 100 RPS separately; the profiles are not implicitly added into a 200 RPS target.

Category	Requests/s	Examples / prerequisites
Catalogue/provider/stock discovery	20	Private item/tariff and public pharmacy/service discovery, price entitlement; cache-hit and miss cases

Category	Requests/s	Examples / prerequisites
Patient/clinical reads	15	Authorized private/direct patient summary, timeline and report metadata, without requiring patient app login
Staff worklists/dashboard	20	Doctor appointments, direct/app lab orders, hospital stores and business/vendor worklists; trusted branch/location filters
Bookings and appointment changes	10	App/walk-in capacity holds and changes; synthetic available capacity
Sales/service billing and POS commands	10	Pharmacy checkout, private consultation/test invoices, counter sales and permitted fulfilment; source/tariff snapshots
Purchasing and stock/consumption commands	10	Receive, reserve/consume, waste, count and issue/receive transfer; pharmacy versus non-medicine item/UOM rules
Collection verification/receipts	5	Stubbed gateway verification and connected cash receipt recording
Inbound provider events	5	Authenticated events with deliberate replay in a separately tagged subcase
Follow-up/service updates	5	Task completion, lab transitions or vendor order updates
Total	**100**	Multi-request journeys must allocate each request within these rates

Some categories refer to proposed APIs. Do not build a benchmark around invented endpoint paths before contracts exist. Baseline current endpoints first and report the reduced scope explicitly; rerun the full mix when the missing modules ship.

Suggested starting dataset, to be adjusted to actual first-year projections: 100 organizations with multiple branches and staff roles; 100,000 synthetic patients including shared caregiver phones/no-phone/dependents; 500,000 historical bookings and 500,000 pharmacy orders; 5,000 medicines and 100,000 pharmacy/medicine stock records with a realistic batch distribution; released/amended report metadata and pending care tasks. Include aged data, busy organizations, sparse filters and hot medicines. Avoid unrepresentative tests consisting only of fresh indexed records in a tiny database. Log actual counts and bytes.

Include both lab partner modes in that dataset. Demonstrate an API-supplied report and a panel-uploaded report creating one release/review outcome each, repeated-click/callback safety, cross-lab/branch denials, and explicit handover without a second booking. Report upload byte throughput remains a separate test; panel worklist and status commands remain in the 100 RPS mix.

Add representative lab reagents/kits/consumables, clinic supplies and hospital department locations, including fractional stock units, lots/expiry and in-transit transfers. Seed private tariffs, direct patients without app accounts, supplier purchases, opening balances and software-plan records. Test publication off/on, unauthorized private discovery, concurrent stock consumption/sales, transfer replay and financial source attribution. A test/procedure/report retry must not duplicate consumption, receipts or commission; business invoices and platform subscription/settlement records reconcile separately.

Keep normal JSON responses bounded, for example a documented maximum page size and a representative 10-50 KiB response distribution. Test 1-10 MiB uploads/downloads separately using private object storage with authorization; document concurrency and MB/s. Large reports and exports must not silently share the same tiny-payload latency claim.

Architecture starting point

Retain the modular backend. Start staging with two independently restartable API instances, one dedicated worker process with a supervised spare/recovery strategy, a managed MongoDB replica set, private object storage and a TLS load balancer/reverse proxy. Place components across failure domains if availability requires it. A two-process deployment on one VM does not provide host-failure resilience.

For an initial experiment, allocate roughly 2 vCPU/4 GiB per API process and worker, and an appropriately sized database tier with measured IOPS/memory; this is a benchmark starting point, not a production recommendation without results and cost review. Document actual hardware, region, network, DB topology/storage, container limits, runtime versions and worker concurrency. Redis is optional: introduce it for a selected queue/cache need after comparing it with hardening the existing persisted Mongo queue.

Budget connections across every process. Example starting ceiling: two API pools of 20 plus worker pool of 10, with additional space for monitoring, migrations, standby processes and failover. Tune from wait-time and server limits rather than multiplying default pools as replicas increase. Set bounded server-selection/socket/request timeouts and backpressure; never allow unlimited provider calls, queues or file parsing in API memory.

Before multi-instance API deployment:

1. Remove local-file ownership assumptions; all clinical documents need shared private storage and audited access.
2. Move startup backfills/index changes into controlled deployment jobs. Existing `syncIndexes()` can drop undeclared indexes and must be reviewed.
3. Move cron/consumer ownership out of API startup or use a tested leader/lease strategy. Ensure timeout and cleanup jobs remain idempotent with overlapping workers.
4. Add liveness/readiness, in-flight drain and SIGTERM shutdown for HTTP, workers and DB connections. Readiness must remain false until required dependencies initialize.
5. Add ingress/auth/provider-client rate limits, request-size limits and timeouts. A fast 429 response to useful offered traffic is not a 100 RPS capacity pass.

Code-specific bottlenecks to investigate first

Existing path / observation	Experiment and smallest appropriate change
`file/controller.serveFileByPath`: synchronous disk read/public cache	Move private data off public static serving; use controlled object access/streaming and measure event-loop delay and transfer memory
`specialistPortal.generatePrescriptionPdf`: local filesystem and external `wkhtmltopdf` per request	Dedicated bounded document jobs; retain pending/failure status and version identity, measure PDF throughput separately
`pharmacyOrders.resolveOrderItemsPricing`: sequential inventory and medicine lookups per line	Profile query counts; fetch needed medicines/inventory in bounded batches and preserve authoritative pricing/availability checks
`externalBookings.syncBookingFromWebhook`: report/provider/email work on callback path	Persist verified inbox first, process with worker, track processing failure and safe replay
`TaskQueue`/`JobConsumer`: one claim per init, ten-second scheduler, inline publish consumption	Measure queue growth; add bounded draining, independent worker concurrency and retry/lease recovery before adding queue technology
Large service classes and provider fan-out in bookings/assessments	Trace dependency latency and cap provider concurrency/timeouts; cache only safe catalogue data with a defined staleness policy
Discovery/summary queries and embedded batches	Run `explain('executionStats')` on representative filters, sorting and tenant scope; measure geo queries and hot-stock write contention before selecting indexes/data changes
Root request logging/startup behavior	Use structured sampled request logs without PHI and remove per-request console noise; readiness gates and migrations outside API start

Candidate index patterns must come from the actual query plan: organization/branch + state + created/due date for worklists; patient + encounter time for timeline; provider + external ID for mappings; organization + operation + idempotency key; event/delivery state + next attempt. Some current indexes already exist. Deduplicate declarations and review uniqueness/partial filters before changes. Add a geospatial representation/index only if measured discovery queries warrant it; the current latitude/longitude fields alone do not establish an indexed spherical search.

Catalogue caches may tolerate bounded staleness; checkout revalidates price/stock. Do not use a shared cache as the authority for balances, reservations, clinical access grants or report release. Keep clinical response caches private and permission-aware.

Test stages and evidence

1. **Reproducible build and functional gate:** repair lock/install, type/build checks, real replica-set integration tests, seeded test organizations and role tokens. Record all skipped cases; no production credentials or live payment actions.
2. **Single-instance baseline:** use current APIs and controlled provider stubs, profile DB plans/queries, event-loop delay, CPU/memory and queue age. Fix correctness before tuning throughput.
3. **Ramped steady test:** 5-minute warm-up, short 25/50/75 RPS steps, then **30 minutes at 100 RPS**. At the target, approximately 180,000 measured requests should be scheduled. Retain per-endpoint/role/organization results and business reconciliations.
4. **Soak:** 2 hours at 100 RPS after the steady gate passes, about 720,000 scheduled requests. Include normal task/import/report worker traffic and check memory, storage and backlog trends.
5. **Contention/correctness:** at least 100 concurrent attempts for last stock/appointment capacity; mixed sale/procedure consumption; transfer dispatch/receipt replay; fractional UOM conversion; invoice/receipt/refund retries; private/public

access; reordered/amended reports. Assert exact quantities/capacity, source/commission and business/platform ledger balances. These races do not replace the useful-work mix.

6. **Recovery:** stop one API, restart a worker during provider delivery, delay provider responses and simulate a failed storage/DB connection. Reconcile accepted writes and queue replay. Run cluster failover only in an isolated authorized staging environment.
7. **Provider contract:** low-rate sandbox checks for LabStack, Healthians, Razorpay and notification services with agreed payloads/signatures/quotas/timeouts. Report these independently from stubbed throughput. Replay duplicate/stale events and validate allowed state changes. Include authorization without capture, capture/expiry and cancellation races, bad then valid client signature, unknown gateway-order creation, second captures, preliminary/final/corrected reports and explicit wallet-credit versus original-method refund reconciliation.
8. **Optional headroom test:** briefly increase to 150-200 RPS to observe controlled overload/backpressure and recovery. This characterizes limits; it does not change the 100 RPS release target.

Use k6 or an equivalent arrival-rate tool. In k6, an arrival rate sets **iterations**, not intrinsically HTTP requests/s. Allocate one measured request per iteration or explicitly calculate aggregate request counts across scenarios, including retries. Supply enough VUs for measured latency and fail the run on dropped iterations. Official constant-arrival-rate documentation.

Each run must retain: application commit/config, dataset seed/counts, hardware/topology, test script revision, provider mode, offered/started/completed/successful rate, p50/p95/p99 per endpoint, failures/timeouts/conflicts, dropped work, DB explain/profiling summary, CPU/heap/event-loop/connection/queue metrics, stock/slot/money reconciliation and exact pass/fail reasons. Do not attach patient data or secrets to results.

Wallet/payment contention fixtures include two concurrent spends against funds for one, duplicate and distinct partial refunds, crash between balance/movement writes, capture then failed client verification then Pay Now, and authorized/unknown-attempt retry. Require no duplicate gateway order or wallet effect, no negative balance and exact movement reconciliation. Test finance cutover with known opening balances and post-cutoff replay before M2 posting; these are not executed results.

Cross-destination refund fixtures race a full wallet credit against a gateway/cash refund on one captured payment. They must contend on one guarded source budget, with exactly one full claim; pending/unknown external refunds retain reserved headroom through timeout/restart and partial returns never exceed the source total.

F14 adds a separate basic-phone/provider trial: simultaneous top-ups against limited partner funding, app/phone balance and slot races, shared-number/PIN denial, missing/duplicate/reordered call events, no-answer, reconnect and uncertain completion. Agree concurrent-call count, setup/queue/reconnect SLO, consent/access checks and per-consultation cost before the live pilot. The canonical 100 HTTP RPS profile is unchanged and does not establish simultaneous-call capacity; tag phone control/callback tests separately and rebase any future representative workload explicitly. Phone acceptance.

F14 security/money/clinical fixtures add a synthetic PIN canary across provider/local logs and persistence, a database-only copy without its separate key, a 1,000-reference spray within the configured abuse window across replicas, Partner-paid consultation server-price/underpayment/duplicate commands, and an ordinary debit racing active held funds. Validate selected funding-purpose restrictions, mutually exclusive hold capture/release, actual-mode/eligible-follow-up prescribing and referral-only charge/refund outcomes. These are unexecuted scenarios and do not change the canonical workload.

Cumulative F14 acceptance additionally covers the original basic-phone/no-SMS, funded-top-up race, shared-phone/caregiver denial, queue/shared doctor capacity, one-charge/reconnect, authorized outcome and callback-replay journey. Test private enrolment with a copied slip failing without independent contact proof; single-use/expired/concurrent challenge consumption; setup/reset/issuance/recovery throttling; completed-change notification and replay revocation. A reset request alone must not lock out a valid patient.

Under a defined cross-account spray at tested capacity, staged controls must throttle abusive traffic while a legitimate verified-contact caller completes PIN verification through reserved capacity; keep authorized encounters and urgent/assisted routes available without bypassing identity/spending checks. Add same/changed practitioner, prior in-person versus remote-only encounter, six-month boundary and changed-condition/context fixtures. These are unexecuted acceptance requirements; global throttling is not a blanket availability guarantee.

Reliability and release ownership

Proposed recovery starting objectives are **RPO <= 15 minutes and RTO <= 2 hours**, subject to clinical/operations and budget approval. They are not existing guarantees. Back up encrypted database and object data with integrity/version metadata, align retention/legal holds, and restore into a clean isolated environment. Check patient/document links, invoice/ledger totals and pending event recovery. After restoring, re-pull gateway and lab-provider facts across an overlapping recovery window; reconcile captures, wallet liabilities, business dues and report versions newer than the backup without duplicate effects. A successful backup job without a demonstrated restore does not satisfy the gate.

DevOps owns deployment, backup/restore and metrics; backend owns transactional correctness and bounded dependency behavior; QA owns the repeatable harness and evidence; clinical/operations own urgent escalation and service-continuity procedures. Release only when the agreed functional and performance gates pass together. If one instance meets the target, extra services/databases are unnecessary; if it does not, fix measured bottlenecks and remeasure before a technology migration.

F14 contact-binding abuse acceptance (AR-25): include unverified contact possession probes, setup/recovery/protected-lane denials, cashier-owned-number self-approval attempts, shared-family exceptions and concurrent contact-version changes in the separately sized phone workload. Staff-number/unrelated-patient reuse checks and key outage/rotation/compromise-reset drills need bounded queues and named P3 approval; unauthenticated proposals cannot lock out valid patients. These are unexecuted scenarios and do not alter the canonical 100 HTTP RPS table.

v1.7 restore acceptance: system15.1-15.2/D9 distinguish generic RPO/RTO from recovery of acknowledged cash, dispensing, manual returns and clinical finalization. Test post-backup revocation/challenge consumption, fresh-login attempts against stale restored authority, and replay of local irreversible effects with independently recoverable evidence or explicit frozen exceptions. A database-local outbox or provider re-pull alone is insufficient. Choose/test the protected recovery generation, per-effect evidence and admission runbook before pilot; no zero-loss or zero-RPO claim is made. The canonical HTTP workload and thresholds remain unchanged.

7. Open source reuse assessment

Source: documentation/planning/OPEN_SOURCE_OPTIONS.md

Research date: **8 November 2026**. **35 distinct projects screened**, including exclusions because their current terms or maintenance do not fit. This is a bounded comparison, not an inventory of every public repository. No candidate was installed, executed, imported or merged. **100 RPS remains unverified.**

The integration baseline is Jansetu's Express 4, TypeScript 5.4, Mongoose 6.6, TypeDI/Joi, Razorpay, provider adapters and Mongo jobs. The capability matrix identifies existing pharmacy batches/reservations, retail/wholesale prices, connected counter orders, appointments and payments. The business specification adds private daily operations for pharmacies, labs, clinics and hospitals, with optional marketplace listing. Labs may use APIs or a Jansetu panel; counter sales require connectivity initially; hospital OPD/billing precede advanced HMS.

Detailed clinical `(documentation/planning/oss-research/clinical-notes.md)`, commerce `(documentation/planning/oss-research/commerce-notes.md)`, platform `(documentation/planning/oss-research/platform-notes.md)` and supplementary `(documentation/planning/oss-research/supplementary-notes.md)` notes contain inspected source URLs, metadata snapshots and unresolved checks. See the method `(documentation/planning/oss-research/METHOD.md)`, citation registry `(documentation/planning/oss-research/CITATION_REGISTRY.md)` and counter-review `(documentation/planning/oss-research/counter-review.md)`. Upstream features below are documented claims, not proof they exist in Jansetu or satisfy its acceptance criteria.

Recommended decision

Keep this backend. Trial small reusable components first: CASL for server permissions; Refine Core for the missing business panels if the frontend team chooses React, with react-admin as the alternative; k6 as a separate verification tool. Compare hardening Mongo jobs against BullMQ before replacing the queue. These supply primitives rather than completed business modules. The table below links their actual license and integration evidence.

Evaluate ERPNext plus Marley as one separate business/clinical stack when substantial purchasing, stock, accounting and OPD requirements justify it. InvenTree is a narrower supplies alternative; Odoo Community is the alternative ERP. **Evaluate OpenELIS Global for advanced lab operations**, while small labs without APIs can use a native Jansetu panel. API absence alone does not require a full LIS.

These are engineering recommendations with **medium confidence** until trials pass. No full application has sufficient evidence for an immediate whole-repository merge. A matching language or database does not establish compatible models, permissions, licenses or transactions. Conversely, an exact-fit hospital/lab system could save considerable workflow work. Compare those savings with adapters, extra databases, reconciliation, migration, training and upgrades. Use the roadmap's post-M0 planning windows and re-estimate M0 at kickoff; no reuse-related schedule saving is assumed before evidence.

What we can actually reuse

Route	Examples	Adoption boundary
Package	CASL; BullMQ if justified	Pin compatible published versions; test actual query/effect paths
Frontend framework	Refine Core or react-admin	A panel app over Jansetu APIs; backend permissions still authoritative
Independent tool	k6	Synthetic workload and actual HTTP metrics; no backend source merge
Separate system/API adapter	ERPNext/Marley, InvenTree, OpenELIS	Explicit record ownership, identifiers, outage recovery and reconciliation
Selected source files	Only after inspection	Exact commit, file license, notices, dependencies and recorded changes; porting retains provenance
Workflow reference	Screens and business processes	Learn the workflow; copying code remains a separate licensing decision

Platform libraries, panels and commerce frameworks

Confidence: **high** for inspected licenses and declared stacks; **medium** for fit; production compatibility unverified. All ten repositories returned unarchived metadata; source commits are recorded in the platform notes. This is not release certification. A small native policy/job implementation could be cheaper than adopting another dependency or service.

Project	License / stack evidence	Recommendation and limit
CASL	TypeScript/JavaScript; MIT; Mongoose adapter manifest	**First backend library trial.** Declared peer range includes our Mongoose 6.6. Prove organization/branch/assignment/field policies across lists, writes, exports, files and aggregations. A peer range is not a passing build.
Refine Core	React/TypeScript; MIT; access provider	**Preferred panel trial.** Data/auth adapters over Jansetu APIs for lab, doctor, stock and billing screens. Clinical signoff and server enforcement remain ours. Commercial Refine services are separate.
react-admin	React/TypeScript; MIT core; paid RBAC	**Alternative to Refine.** Compare the same lab flow; choose one framework. Do not count paid RBAC/scheduler packages as free core features.
BullMQ	TypeScript/Node; MIT core; connections, PostgreSQL	**Conditional queue trial.** Redis-compatible default; current source/docs also describe PostgreSQL. Verify chosen published release contains that backend. Both add infrastructure to Mongo. Outbox delivery/idempotency remain application work; Pro is separate.
OpenFGA	Go service/Node SDK; Apache-2.0; storage	**Defer.** Useful for complex relationship access; adds tuple lifecycle, consistency and availability. Avoid two competing policy authorities.

Project	License / stack evidence	Recommendation and limit
Medusa	Node/TypeScript/PostgreSQL; MIT exceptions, proprietary Enterprise RBAC/SSO	**Reference or deliberate separate commerce replacement.** Own framework and order/stock model; no Express/Mongo drop-in. Prescription release, FEFO and clinical isolation need demonstration. Exclude unlicensed Enterprise paths.
Vendure	NestJS/GraphQL/SQL; GPLv3/commercial, plugin exception	**Later commerce alternative.** Channels are useful but do not prove clinical permissions or settlements. Plugin exception does not make copied framework code MIT.
Novu	Node/Mongo/Redis services; MIT core, Enterprise terms	**Later notification API service.** Compare native outbox/current providers; provider fees, consent and sensitive-data limits remain ours. Enterprise hosting restrictions need separate review.
Chatwoot	Rails/PostgreSQL/Redis; MIT exceptions, Enterprise license	**Later support service.** Conversations/tickets, not the operations panel or clinical record. Define staff scope and attachment retention.
k6	Go CLI/JS scripts; AGPLv3; executor semantics	**Development/CI tool candidate.** Measure HTTP RPS, latency, errors and dropped iterations. Iterations/second need not equal HTTP requests/second. Installation proves no capacity.

ERP, stock, counter and wholesale options

Confidence: **high** for inspected license/source facts; **medium** for workflow fit; adoption cost unverified. These nine projects overlap existing stock, pricing, invoices and payments. Completing native purchases/returns may cost less than integration. Snapshots describe default-branch heads, not selected stable releases.

Project	License / stack evidence	Recommendation and limit
ERPNext	Python/Frappe/relational; GPLv3; batches, price rules, REST	**First broad back-office trial with Marley.** Purchasing/stores/accounts fit daily operations. Prove pharmacy release, returns and marketplace attribution. Current develop targets v17 development requirements; pin a compatible supported release pair.
Odoo Community	Python/PostgreSQL; 20.0 LGPLv3/exceptions; public 19.0 expiry and POS	**Alternative ERP/POS trial.** Public expiry/FEFO evidence is stronger than a generic stock claim. Observed default is 20.0; inspected feature evidence is 19.0. Verify selected-version Community/Enterprise/API boundaries and add-on terms.
InvenTree	Python/Django/React/SQL; MIT; tracking, stock schema	**Supplies-engine alternative trial.** Receipts, locations, expiry and traceability suit consumables. Not demonstrated as pharmacy-safe FEFO/POS/billing. Enable/test audit tracking and prove service-linked consumption and business isolation.
Dolibarr	PHP/SQL; GPLv3-or-later/dependencies; TakePOS	**Second-tier ERP integration/reference.** Relevant counter, lot/date and REST workflows. Pharmacy-safe allocation, batch-correct returns and tenant isolation remain unverified.

Project	License / stack evidence	Recommendation and limit
Tryton / upstream	Python/SQL; GPLv3-or-later; stock modules	**Modular ERP alternative/reference.** Lots, warehouses, units and shelf-life modeling; pharmacy FEFO/local billing not established. GitHub is a mirror; inspect canonical upstream.
OpenSourcePOS	PHP/CodeIgniter/MySQL; MIT text plus mandatory unchanged visible footer	**Counter reference/conditional standalone pilot.** Receiving, barcodes, register and cash-up help. Not plain MIT for rebranding. Medical batches, wholesale, sync and tenancy unverified.
uniCenta oPOS	Java desktop; official FAQ says GPLv3; source archive license bundle not inspected	**Hardware/desktop reference; defer.** Official source index is established, official GitHub repo is not. Verify archive provenance, API access and database versions; avoid random mirrors.
IDURAR	Express/Mongo/React; AGPLv3 root, manifest says Fair-code License, Mongoose 8	**Invoice/quote reference; code trial needs license clarification.** Close language fit does not remove auth/schema/version migration. Pharmacy stock and SaaS tenancy unverified.
Akaunting	PHP/Laravel/SQL; current BSL 1.1 production restrictions, later version-specific GPL change	**Exclude current source for this SaaS shortlist.** Inspected user/company/invoice, branding and accounting-service restrictions conflict with assuming unrestricted multi-business hosted reuse. Suitable separate terms could change selection.

Clinical, hospital and laboratory options

Confidence: **high** for inspected licenses except GNU Health's unresolved current-source terms; **medium** for fit; interoperability/security unverified. ERPNext is counted once above. Marley is its companion. These clinical systems are alternatives, not mandatory additions to one deployment. An existing partner may still warrant an adapter to a lower-priority system.

Project	License / stack evidence	Recommendation and limit
Marley	Python/Frappe/ERPNext; GPLv3; portal, appointments	**Trial with ERPNext.** OPD/IPD, lab, procedures and appointment/invoice workflows. v16 portal payment collection needs Payments app setup. Jansetu sharing/commissions/tenancy require proof. Historical frappe/health is the same project's alias.
OpenELIS Global	Java/Spring/React/PostgreSQL; MPL2; features	**Advanced LIS trial when needed.** Sample lifecycle, validation, reports, instruments and inventory claims. Prove actual API/events, corrected reports and private retrieval. Light deployment can omit optional exchange sidecars. Small panel labs can stay native.
Bahmni	Multi-service hospital distribution; wrapper MIT, mixed constituent licenses	**Alternative hospital stack.** OpenMRS/lab/Odoo/imaging breadth adds operating cost. Docker MIT does not license all images; bundled OpenELIS fork terms do not establish current OpenELIS Global terms.
OpenMRS	Java/relational; MPL2 plus healthcare disclaimer; REST	**Alternative clinical authority/partner adapter.** Patients/encounters/observations; needs supported distribution/modules. Core alone is not POS, wholesale, lab commerce or settlements.

Project	License / stack evidence	Recommendation and limit
OpenEMR	PHP/MySQL or MariaDB; GPLv3; API guide	**Practice-management alternative.** Scheduling/EHR/billing/REST/FHIR; US-oriented material is not proof of local billing fit. Map patient and invoice authority explicitly.
SENAITE	Python/Plone core plus LIMS UI; GPLv2; installation	**Lower priority/reference.** Official 2.x guide still requires Python 2.7 and flags lack of security fixes. Recent activity does not resolve runtime support; prove supported path before operating a new service.
GNU Health HIS	Python/Tryton/PostgreSQL; canonical source guide; historical GPLv3, current source-version license not retrieved	**Reference/future alternative.** Hospital/lab scope is relevant. Current license/notices/activity remain open; no import clearance from an old mirror.
Open Hospital	Java/MariaDB desktop/client-server; GPLv3; README marks API/web work in progress	**Hospital workflow reference.** Offline/department ideas help later scope; mature SaaS web panel and API integration need proof.
HospitalRun	Umbrella MIT; TypeScript-related frontend/server	**Exclude new production foundation.** Umbrella, frontend and server archived; frontend/server last-push metadata 2023-01-09. Historical UI ideas do not remove dependency/maintenance cost.

Scheduling, SaaS billing and interoperability

Confidence: **high** for inspected licenses/metadata; **medium** for fit. Seven repositories reported unarchived; this is not a support guarantee. Service payments, software subscriptions, marketplace commissions and vendor payouts remain distinct. A named integration requirement could justify these earlier; current OPD/counter/panel scope does not establish that dependency.

Project	License / stack evidence	Recommendation and limit
Cal.diy	MIT; TypeScript/Next.js/Prisma/PostgreSQL; official Cal.com split	**Scheduling reference.** Commercial production code moved private. Public Cal.diy removes organization/team/workflow features; README recommends personal/non-production use. Permissive terms do not prove multi-business clinic suitability.
Easy!Appointments	GPLv3; PHP/MySQL; REST API	**Optional partner adapter.** Otherwise duplicates existing booking/calendar work and availability authority. No direct TS merge.
Lago	AGPLv3; API/front submodules, Rails/PostgreSQL/Redis deployment; edition/payment FAQ	**Defer complex SaaS billing engine.** Separate service if justified. Not a payment processor; Razorpay and necessary Premium features need proof. Root repository language does not describe the entire deployment.
Kill Bill	Apache2; Java/SQL; architecture	**Alternative to Lago.** Prove payment plugins, invoice mapping, entitlements, retry/refund and operating model. Avoid both engines; recurring medicine orders are a different feature.

Project	License / stack evidence	Recommendation and limit
Invoice Ninja	PHP; current Elastic License 2; official explanation	**Exclude under inspected terms.** Hosted/managed-service limitations conflict with assuming unrestricted third-party invoicing SaaS reuse. Suitable separately verified terms would be required.
Medplum	TypeScript/Node/PostgreSQL/Redis; Apache2; open-source scope	**Optional FHIR/platform trial.** Specific SDK/component may be reusable; full server introduces clinical authority and persistence. Validate exact runtime/exchanges. Not an inventory/settlement/consent implementation.
HAPI FHIR	Java libraries/JPA server; Apache2; JPA docs	**Alternative FHIR boundary, later.** Useful for a named exchange/profile; not a Node module or complete HMS. Do not introduce alongside Medplum without a clear responsibility boundary.

What reuse still leaves us to build

This is proposed Jansetu architecture, not an upstream capability claim. Fewer authorities make failures easier to reconcile.

Record	Authority rule
Business/staff/publication	Jansetu membership, branch access, private operations and optional listing; explicit external tenant mapping. For Frappe, decide site-per-business versus a shared-site/company design and prove both isolation and operating cost; company labels are insufficient
Patient/care	Stable Jansetu patient identity; external patient/encounter/specimen mapping; deliberate clinical authority, consent and amendment history
Stock	One writer per store/balance. If ERP owns it, all channels reserve/consume there; Mongo holds a projection. Otherwise export records and prevent competing stock writes
Lab/order	One commercial order contract for API and panel partners; source attribution, assignment, sample/report states and correction versions; explicit LIS accession mapping
Money	Distinct invoice, receipt, attempt, refund, credit, commission, subscription fee and payout records; name the invoice issuer and payment/refund/ledger owner for each flow, map immutable references and reconcile. Gateway collection is not proof of vendor payout
Files/events	Private access, file validation, signoff, corrected reports, authenticated events, durable delivery/replay and recovery; queue adoption supplies none of these policies automatically

Trial order and pass/fail gates

1. **Foundation first.** Resolve documented clean-install/build, access, file and webhook blockers. Pin a supported runtime. Use an isolated development branch with synthetic data. No production migration is authorized by this research.
2. **CASL plus one panel slice.** Two businesses/two branches; receptionist, clinician, technician, signatory and vendor roles. Show lab inbox, sample action, report upload/finalize/private download in Refine. Compare react-admin if needed. Pass negative list/detail/write/export/file tests, build compatibility and save-failure recovery; record actual custom code and paid-feature needs.
3. **Queue recovery comparison.** Same outbox/job contract on hardened Mongo and selected BullMQ release/backend. Kill workers near side effects, replay duplicates, time out providers and restore stores. Require

recoverable observable attempts, fast API acknowledgement and no duplicate financial mutation; choose one queue after measuring operating cost.

4. **One business-engine trial.** ERPNext/Marley versus native completion; InvenTree if supplies-only, Odoo as alternative ERP. Replay partial purchase receipt, box/strip/tablet conversion/rounding, two expiry batches, expired/quarantined exclusion, simultaneous app/counter last-unit sale, branch transfer, wholesale price snapshot, original-batch partial return, supplier return, credit/refund and consumable wastage reversal. Require isolated second business and reconciled totals.
5. **Advanced lab only when warranted.** OpenELIS order/patient mapping, accession, rejection/recollection, reviewer approval, corrected report, private retrieval and consumable effect. Prove actual event authentication, replay and outage handling. Otherwise ship the bounded native panel.
6. **Provenance/migration/load.** Pin commits, licenses/exceptions/notices and complete dependency inventory; review intended distribution/network model; prove exports, upgrade/rollback and restore. Then measure released workload at 100 HTTP RPS including actual external authorities/failure cases and zero integrity violations. No schedule savings before evidence.

Key disputes and limits

* **Workflow breadth versus operating cost:** established ERP/LIS could save development; adapters, upgrades and reconciliation may outweigh that. Compare the same bounded flow against native completion.
* **Public source versus permission:** copyleft, Enterprise and source-available terms differ. Root badges are incomplete. Separate deployment or porting to TypeScript does not erase provenance or automatically settle obligations.
* **Activity versus suitability:** a push may be a translation/profile change. HospitalRun is archived; SENAITE is active with a documented legacy-runtime constraint. Stars and recent commits are not security evidence.
* **Docs versus selected release:** BullMQ PostgreSQL, ERPNext development requirements, Odoo edition/version and Cal.com's public/private split change selection. Inspect the actual version before implementation.

This research did not audit every dependency, certify legal/regulatory fit, run deployments or rehearse migrations. GNU Health current-source license, uniCenta archive/API terms, IDURAR license metadata and selected-release compatibility remain open. Application implementation and any source import require a scoped follow-up task. Preserve the existing security and 100 RPS acceptance gates.